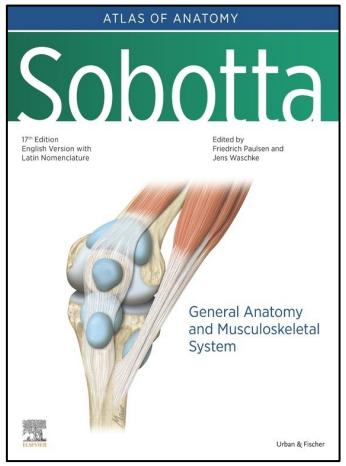
Modul 5.3 Bevægeapparatet

Overekstremitetens (UE) - systematiske anatomi





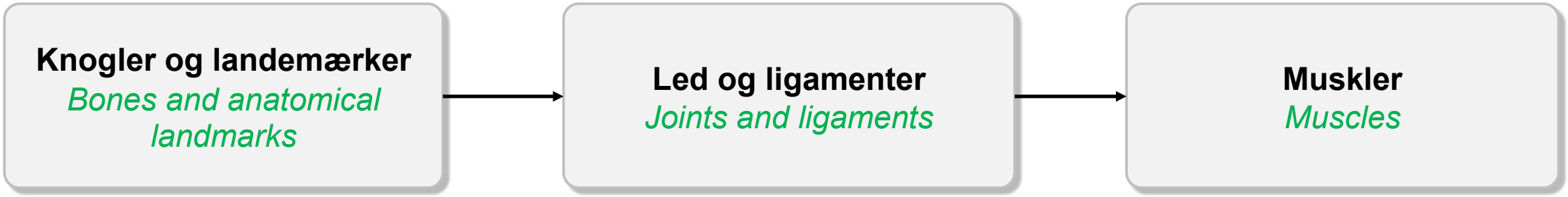
Tong Tong
Assistant Professor

Normal text = Danish
Normal text + Italic = Latin anatomical term
Italic + smaller + green = English

Systematic anatomy

Systematic anatomy gives the basic “parts list” before we study regional relationships.

Regional anatomi: Hvor ligger strukturerne i forhold til hinanden?
Regional anatomy: Where are structures located relative to one another?



Bones provide the framework and palpable landmarks

Joints and ligaments define movement and stability

Muscles create movement and actively stabilize the upper limb

Hvad kan din overekstremitet gøre?

What can your upper limb do?

Enkle hverdagsbevægelser

Simple everyday gestures



Række op
Reach overhead

Vinke
Wave

Pege
Point forward



Knytte hånd
Make a fist



Præcisionsgreb
Precision pinch

Læringsmål:

- **Redegøre for de vigtigste knogler og anatomiske landemærker i overekstremiteten**
  Describe the major bones and anatomical landmarks of the upper extremity

- **Redegøre for de vigtigste led og ligamenter samt forklare deres betydning for bevægelse og stabilitet**
  Describe the major joints and ligaments and explain their role in movement and stability

- **Redegøre for de vigtigste muskelgrupper omkring skulderbæltet og skulderleddet samt forklare deres rolle i bevægelse og stabilisering**
  Describe the major muscle groups of the shoulder girdle and shoulder joint and explain their role in movement and stabilization

- **Redegøre for de vigtigste muskelgrupper i overarm og underarm og relatere deres anatomiske placering til deres funktion**
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- **Redegøre for de vigtigste strukturer i håndled og hånd, herunder fleksorsener, canalis carpi og funktionelt vigtige håndmuskler**
  Describe the major structures of the wrist and hand, including the flexor tendons, carpal tunnel and functionally important hand muscles

Overekstremitetens knogler og ledforbindelser

Bones and joints of the upper extremity

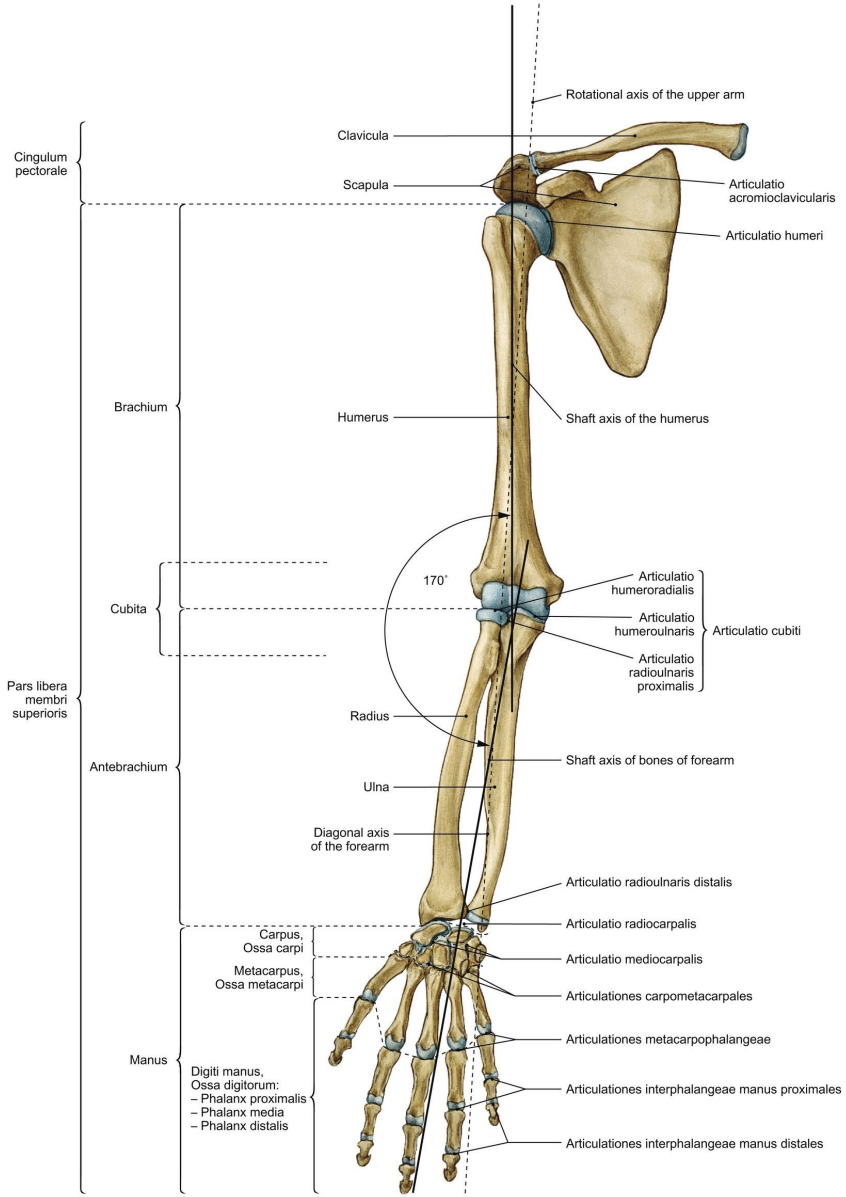
Hvad danner rammen? Overekstremiteten (*Membrum superius* *Upper extremity*)

- Skulderbæltet (*Cingulum pectorale* *Shoulder girdle*)
  - *Clavicula* *Clavicle*
  - *Scapula* *Scapula / shoulder blade*

- Overarm (*Brachium* *Arm*)
  - *Humerus* *Humerus*

- Underarm (*Antebrachium* *Forearm*)
  - *Radius* *Radius*
  - *Ulna* *Ulna*

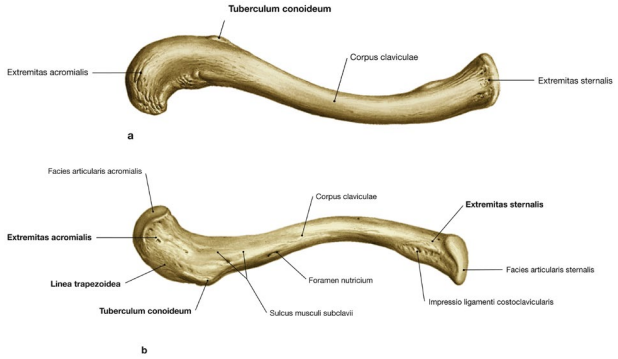
- Hånd (*Manus* *Hand*)
  - *Carpus, Ossa carpi* *carpal bones*
  - *Metacarpus, Ossa metacarpi* *metacarpal bones*
  - *Ossa digitorum* *finger bones*

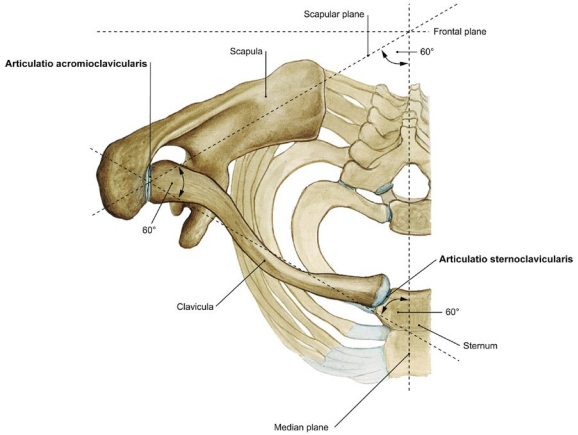


Skulderbæltet: den mobile platform (*Cingulum pectorale*)

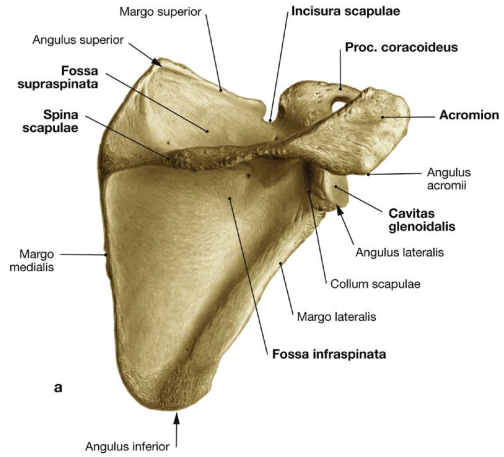
The shoulder girdle: the mobile platform

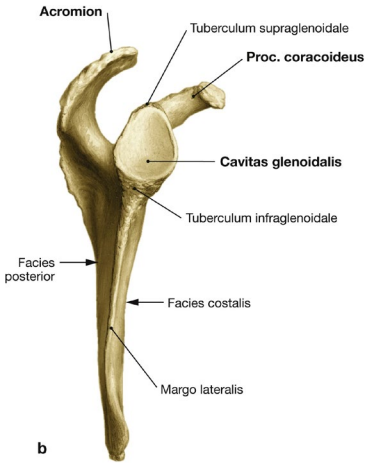
- Kraveben (*Clavicula, Clavicle*)
  - *Extremitas sternalis*
  - *Extremitas acromialis*

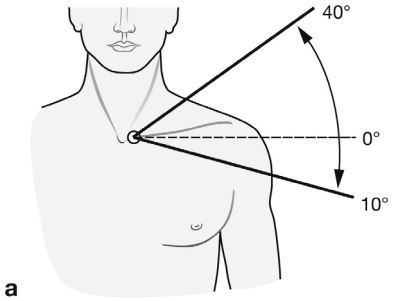




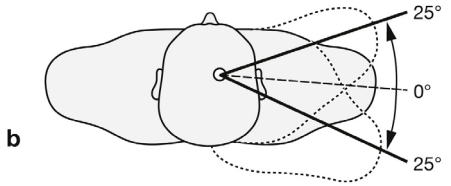
- Skulderblad (*Scapula, Scapula/shoulder blade*)
  - *Acromion*
  - *Processus coracoideus*
  - *Spina scapulae*
  - *Cavitas glenoidalis*







- Scapula fungerer som en bevægelig platform for overarmen.
  The scapula provides a mobile platform for the arm.



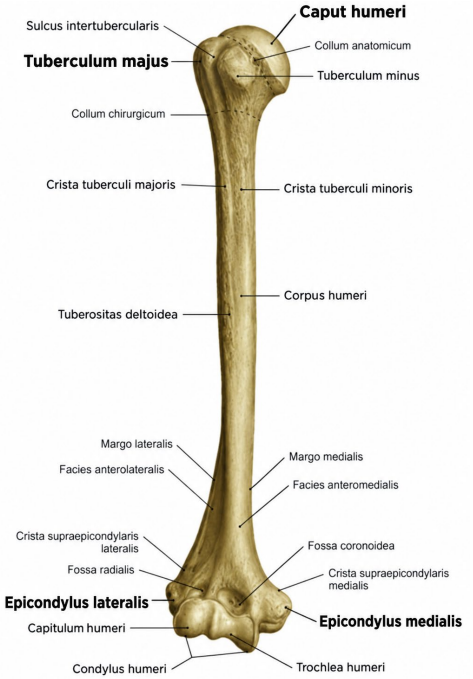
Overarm, underarm og hånd

Arm, forearm and hand

- Overarm (*Brachium, Arm*)
  Humerus *Humerus*
  - *Caput humeri* *Head of the humerus*
  - *Tuberculum majus* *Greater tubercle*
  - *Epicondylus medialis* *Medial epicondyle*
  - *Epicondylus lateralis* *Lateral epicondyle*

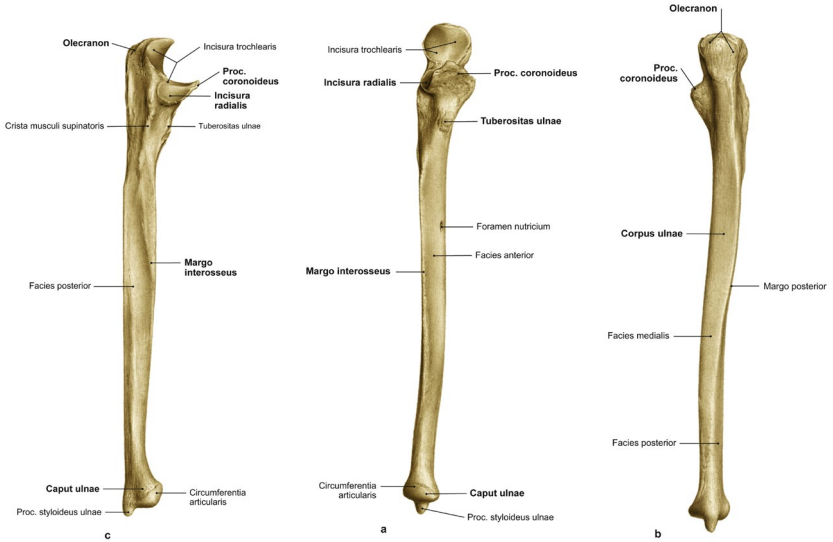
- Underarm (*Antebrachium, Forearm*)
  - *Radius* — tommelfingersiden (*thumb side*)
  - *Ulna* — lillefingersiden (*little-finger side*)

Humerus

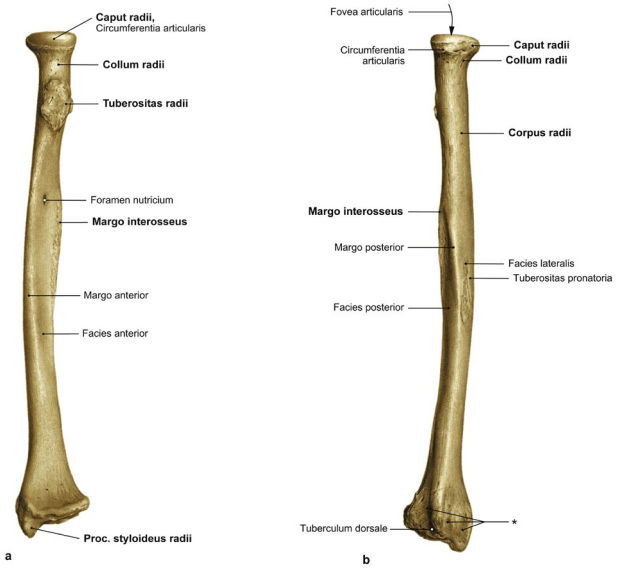




Ulna



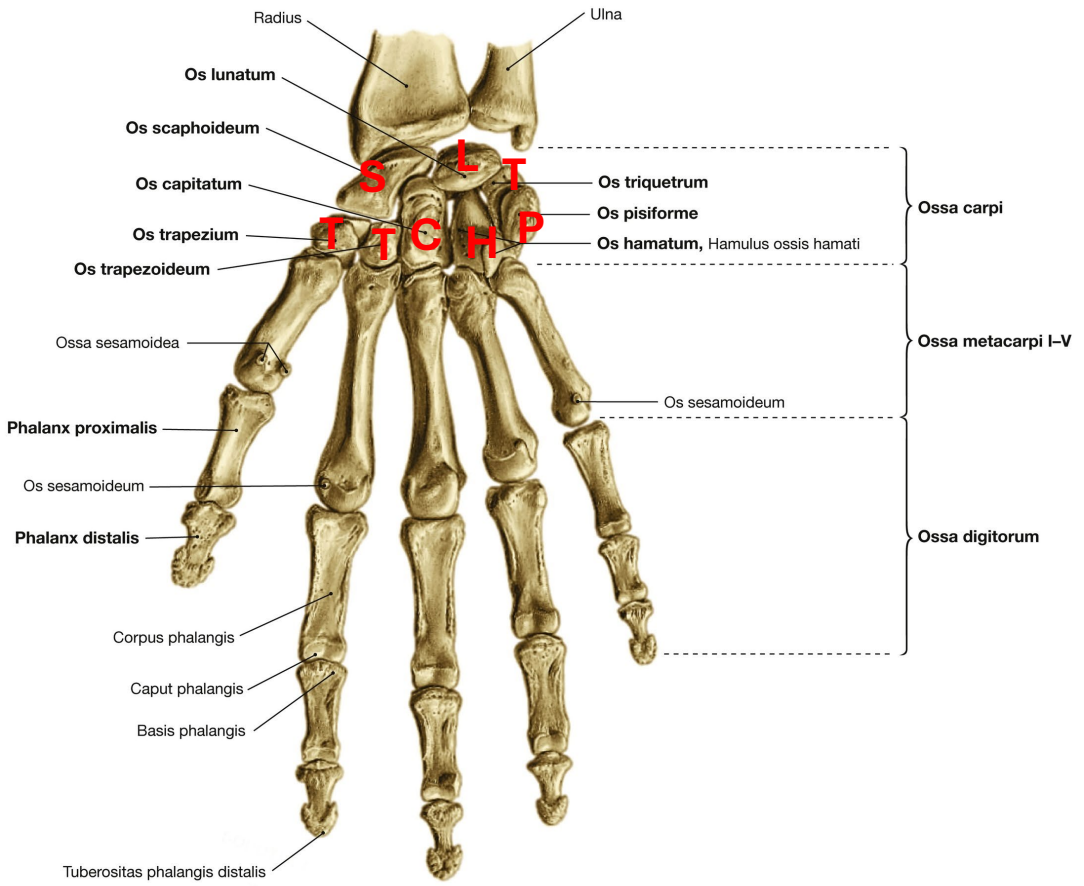
Radius

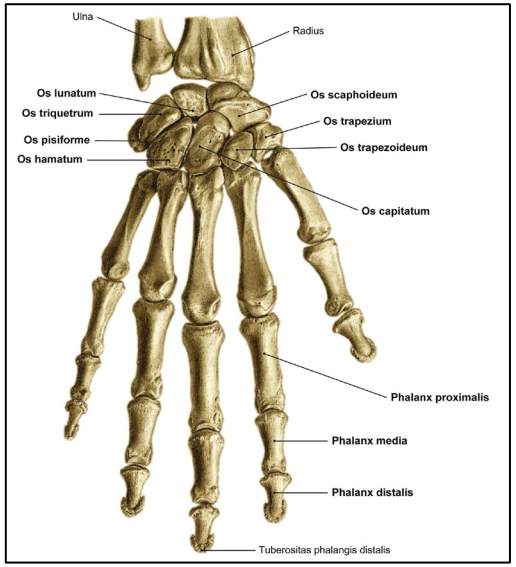


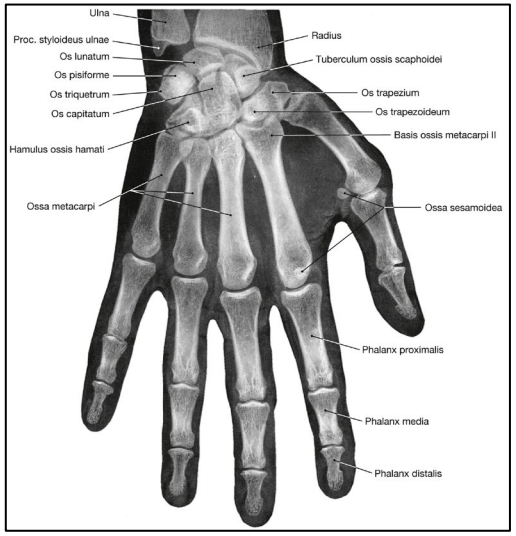
Overarm, underarm og hånd

Arm, forearm and hand

- Hånd (*Manus, Hand*)
  - *Carpus*
  - *Metacarpus*
  - *Digiti manus*
    - *Phalanx proximalis*
    - *Phalanx media*
    - *Phalanx distalis*







En morsom huskeregel for håndrodsknoglerne:

Some **L**overs **T**ry **P**ositions
That **T**hey **C**an't **H**andle

Læringsmål:

- **Redegøre for de vigtigste knogler og anatomiske landemærker i overekstremiteten**
  Describe the major bones and anatomical landmarks of the upper extremity

- **Redegøre for de vigtigste led og ligamenter samt forklare deres betydning for bevægelse og stabilitet**
  Describe the major joints and ligaments and explain their role in movement and stability

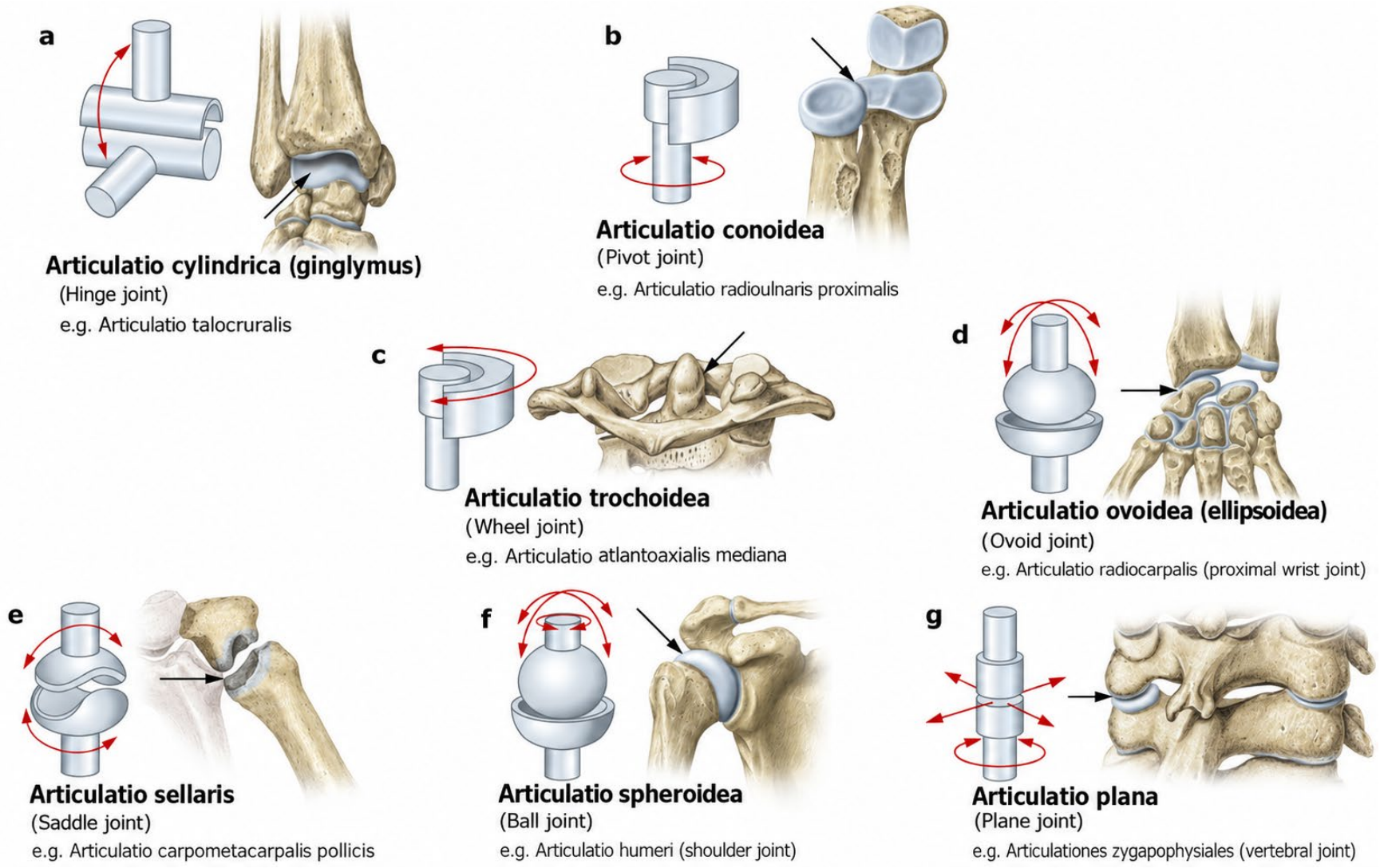
- **Redegøre for de vigtigste muskelgrupper omkring skulderbæltet og skulderleddet samt forklare deres rolle i bevægelse og stabilisering**
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Led i overekstremiteten

Joints of the upper limb



Led i overekstremiteten

Joints of the upper limb

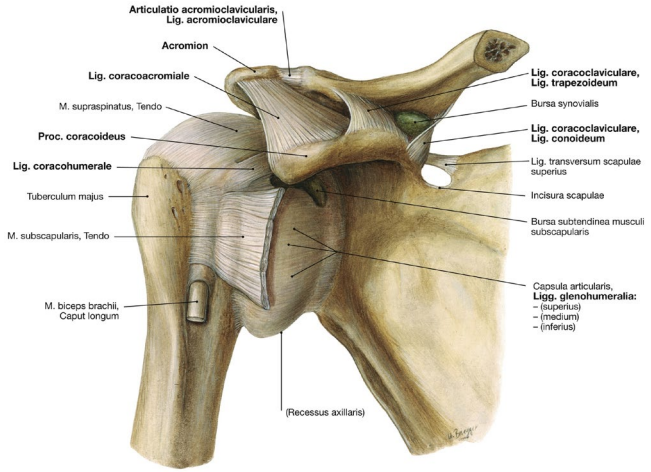
Overview of joints of the upper limb -

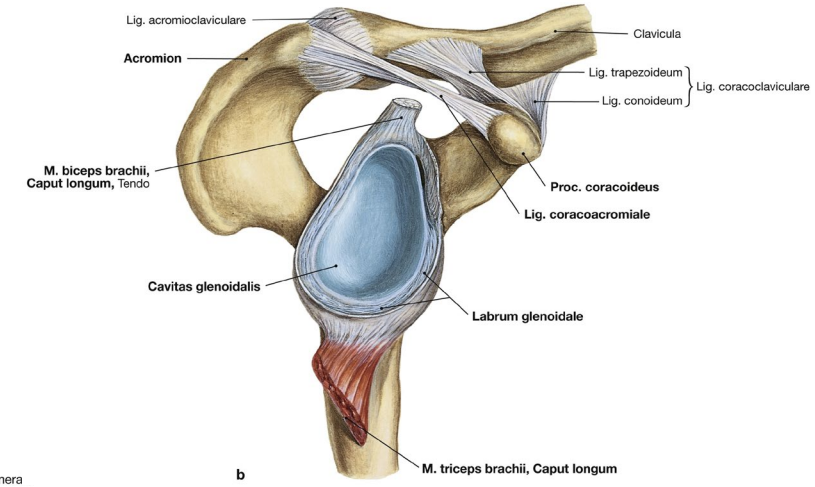
• Medial clavicular joint (**sternoclavicular joint**): ball-and-socket joint	• Proximal wrist joint (**Articulatio radiocarpalis**): condyloid joint
• Lateral clavicular joint (**Articulatio acromioclavicularis**): plane joint (acts along with the medial joint as ball-and-socket joint)	• Distal wrist joint (**Articulatio mediocarpalis**): interlocked hinge joint (along with the proximal joint, functions as condyloid joint)
• Shoulder joint (**Articulatio humeri**): ball-and-socket joint	• Wrist (or carpal) and metacarpal joints (**Articulationes intercarpales, Articulationes carpometacarpales, Articulationes intermetacarpales**): amphiarthroses, except for the carpometacarpal joint of the thumb (**Articulatio carpometacarpalis pollicis**): saddle joint
• Elbow joint (**Articulatio cubiti**): composite joint, consists of: – Humeroulnar joint (**Articulatio humeroulnaris**): hinge joint – Humeroradial joint (**Articulatio humeroradialis**): ball-and-socket joint – Proximal radioulnar joint (**Articulatio radioulnaris proximalis**): pivot joint	• Metacarpophalangeal joints (**Articulationes metacarpophalangeae**): ball-and-socket joints, except for metacarpophalangeal joint of the thumb: hinge joint
• Distal radioulnar joint (**Articulatio radioulnaris distalis**): pivot joint	• Middle and distal finger joints (**Articulationes interphalangeae manus**): hinge joints

Skulderkomplekset
Shoulder complex

- Skulderleddet er et kugleled.
  The shoulder joint is a ball-and-socket joint.

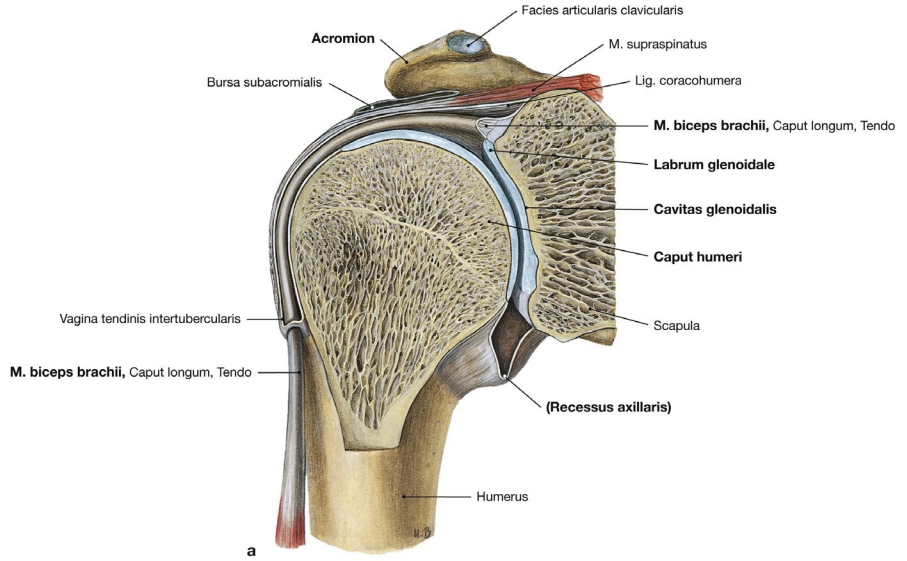
- Vigtige led:
  - *Articulatio sternoclavicularis*
  - *Articulatio acromioclavicularis*
  - *Articulatio humeri*





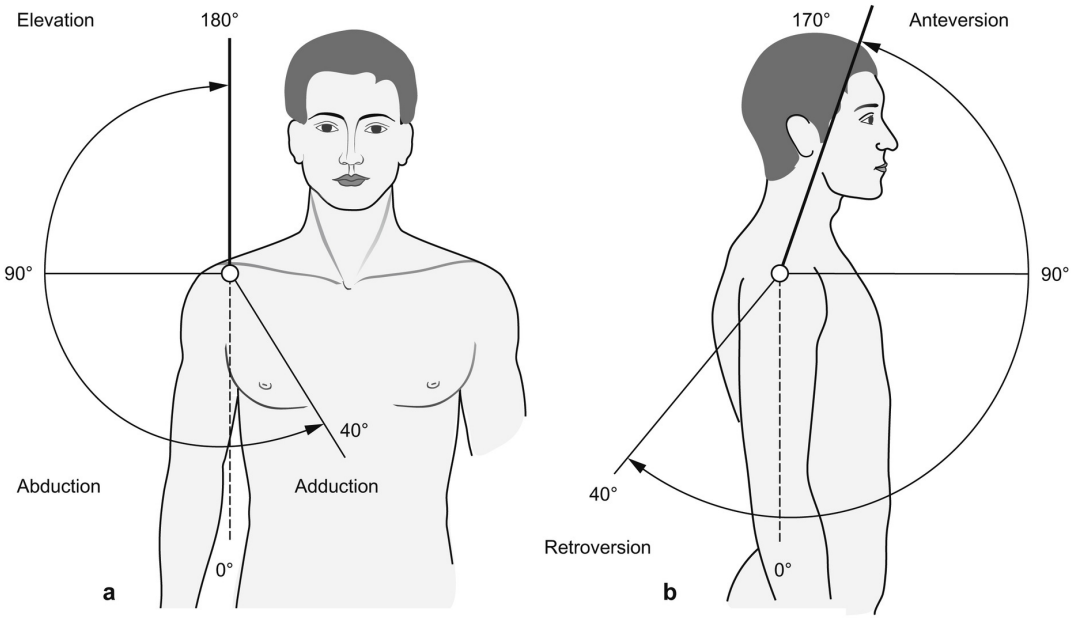
Bevægelse af armen over hovedet kræver både bevægelse af humerus og scapula.
Overhead movement requires motion of both the humerus and the scapula.

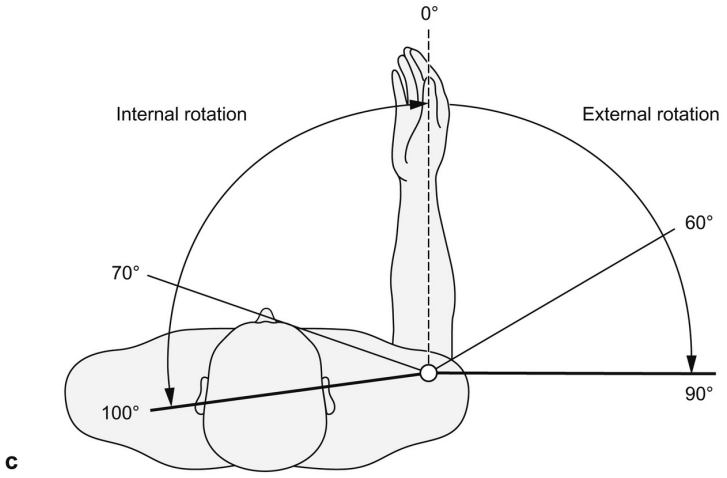
Stor bevægelighed
High mobility
↓
Relativt lav knoglekongruens
Relatively low bony congruence
↓
Større krav til kapsel, ligamenter og muskler
Greater dependence on capsule, ligaments and muscles
↓
Større risiko for instabilitet og luksation
Greater risk of instability and dislocation



Bevægeudslag i skulderleddet med og uden involvering af kravebensleddene.

Range of motion in the shoulder joint with and without the involvement of the clavicular joints.





Range of motion in the shoulder joint alone:

- Abduction–adduction: 90°–0°–40°
- Anteversion–retroversion: 90°–0°–40°
- External rotation–internal rotation: 60°–0°–70°

Range of motion in the shoulder joint with clavicular joints:

- Abduction–adduction: 180°–0°–40°
- Anteversion–retroversion: 170°–0°–40°
- External rotation–internal rotation: 90°–0°–100°

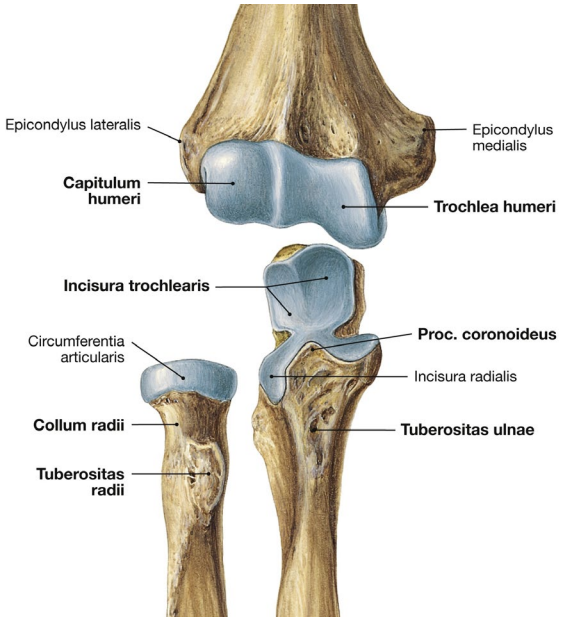
Albue
Elbow

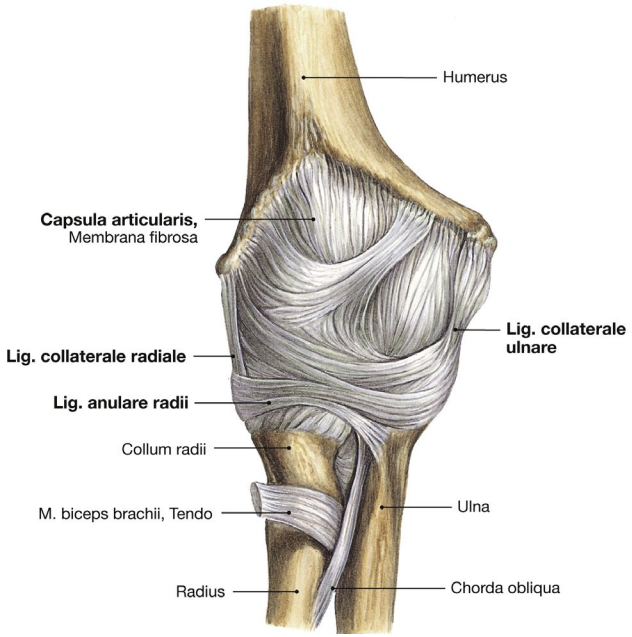
Albueled (*Articulatio cubiti, Elbow joint*)

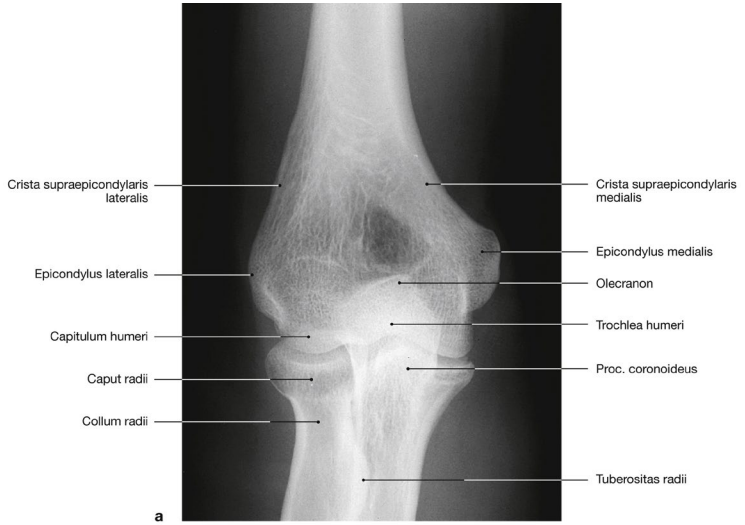
- Primære led:
  - *Articulatio humeroulnaris*
  - *Articulatio humeroradialis*
  - *Articulatio radioulnaris proximalis*

- Primære bevægelser:
  - Fleksion *Flexion*
  - Ekstension *Extension*
  - *Pronatio* *Pronation*
  - *Supinatio* *Supination*

Ventral view

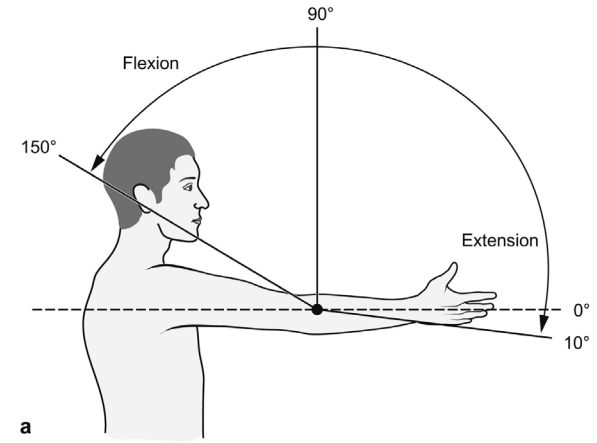


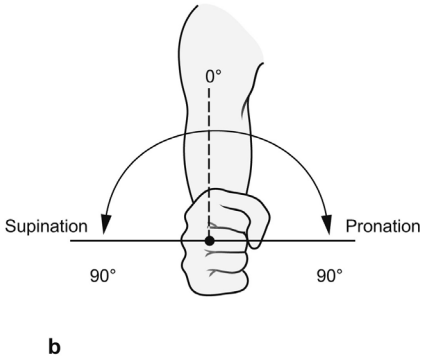


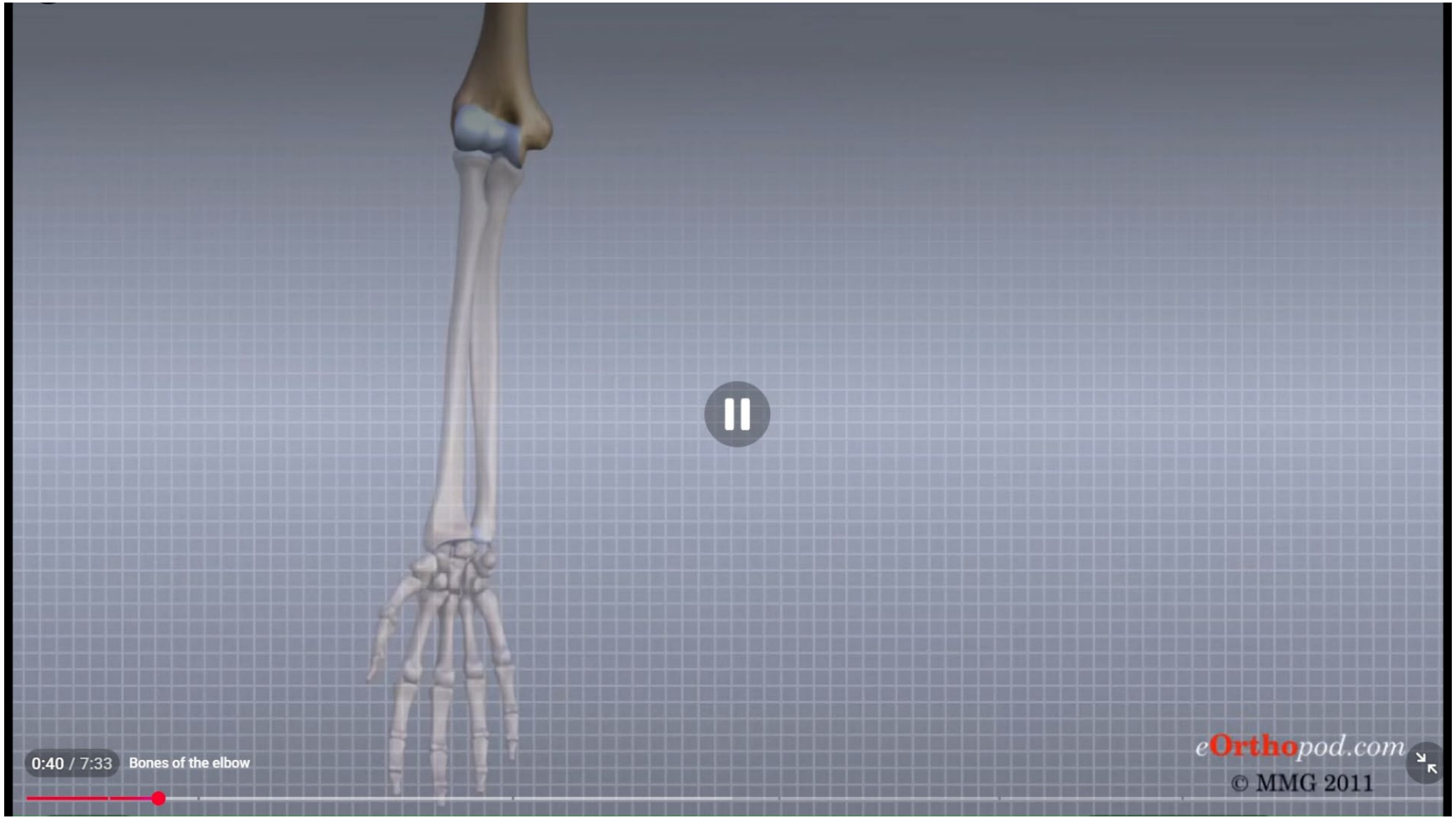


Albue

Elbow


90°
Flexion
150°
Extension
0°
10°
a


0°
Supination
Pronation
90°
90°
b


0:40 / 7:33
Bones of the elbow
eOrthopod.com
© MMG 2011

Håndled
wrist

Articulatio radiocarpalis

“Det egentlige håndled” – ægte sammensat ægled
The “true wrist joint” – a compound condyloid/ellipsoid synovial joint

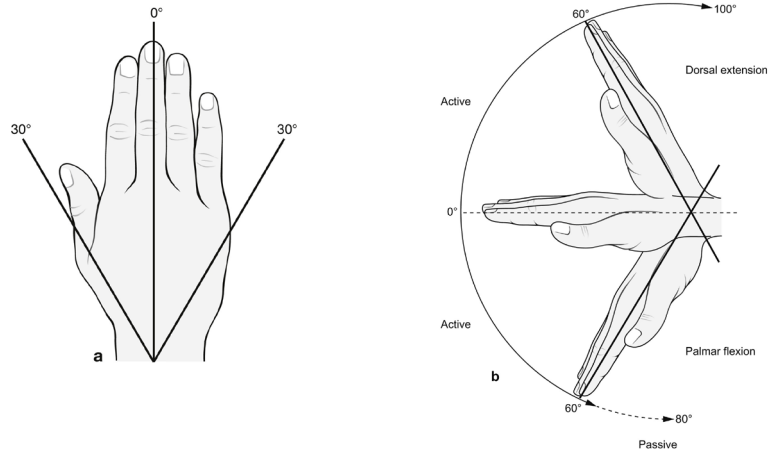
Ledskål
Articular socket

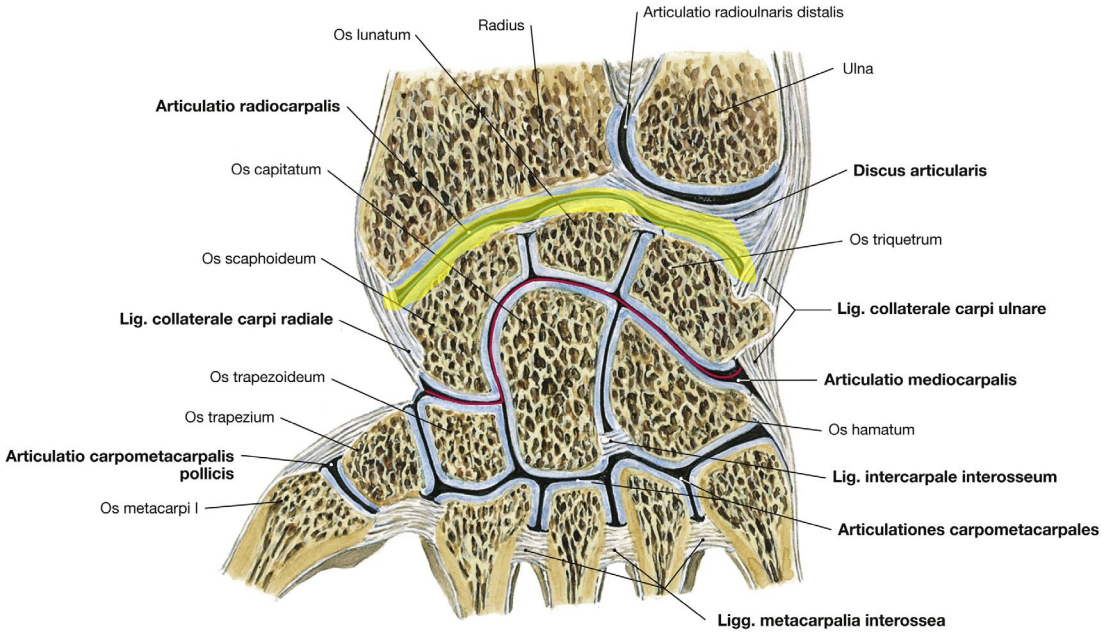
- *Facies articularis radii*
  Articular surface of the radius

- *Discus articularis* – **IKKE ULNA**
  Articular disc – NOT the ulna

Ledhoved
Articular head
- *Ossa carpi* 1–3 og ligamenter
  Carpal bones 1–3 and ligaments

Ulna indgår ikke direkte i *articulatio radiocarpalis*.
The ulna does not articulate directly with the carpal bones in the radiocarpal joint.





Håndled
wrist

Articulatio radiocarpalis

Ledlinjen
Joint line

Proksimal konveks bue mellem de to *processus styloidei*
Proximal convex curve between the two styloid processes

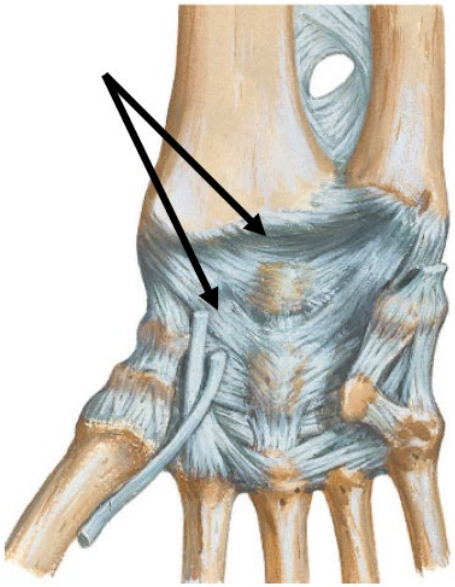
Ligamenter
Ligaments

Ligamenta radiocarpalia palmaria et dorsalia
Palmar and dorsal radiocarpal ligaments

Klinik: Colles-fraktur
Clinical relevance: Colles' fracture

- Fraktur af den distale radius, typisk efter fald på strakt hånd
  Fracture of the distal radius, typically after a fall on an outstretched hand

- Den distale knoglefragment forskydes typisk dorsalt
  The distal fragment is typically displaced dorsally







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Hvad skaber bevægelse?

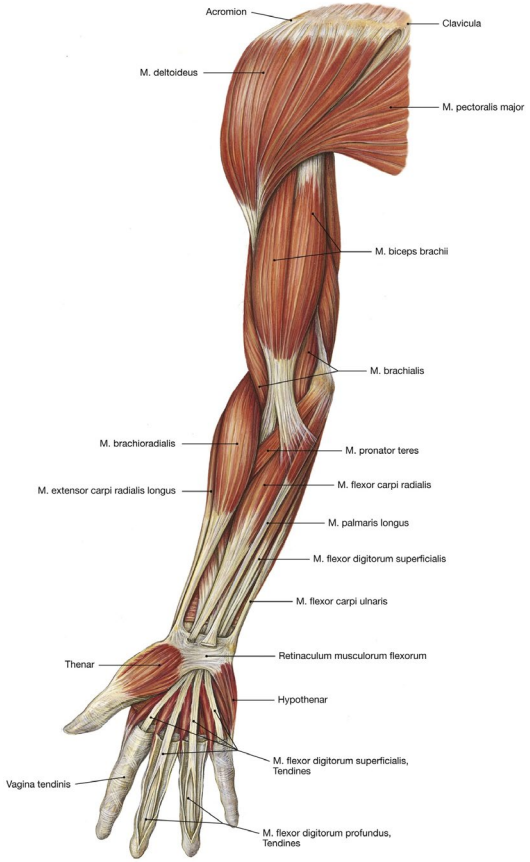
What generates movement?

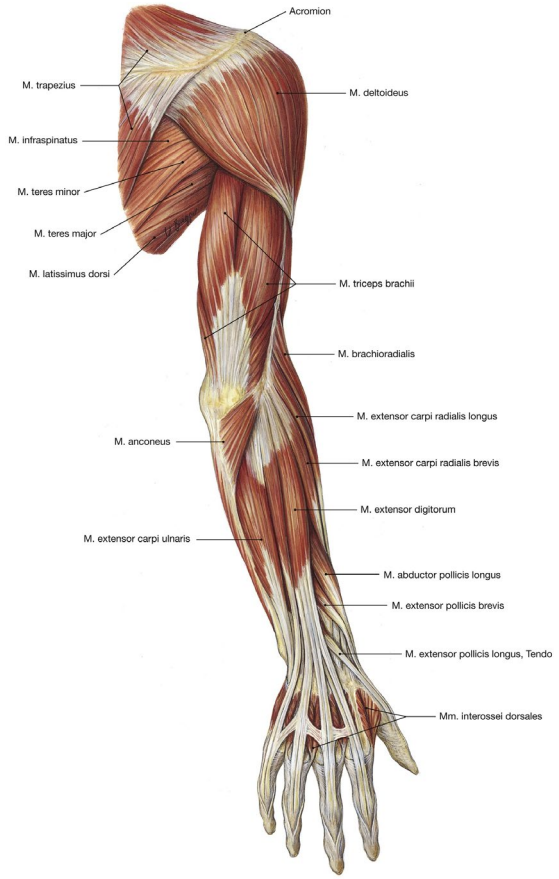
Hvis du ved, hvor en muskel ligger, og hvilket led den krydser, kan du ofte forudsige dens funktion.

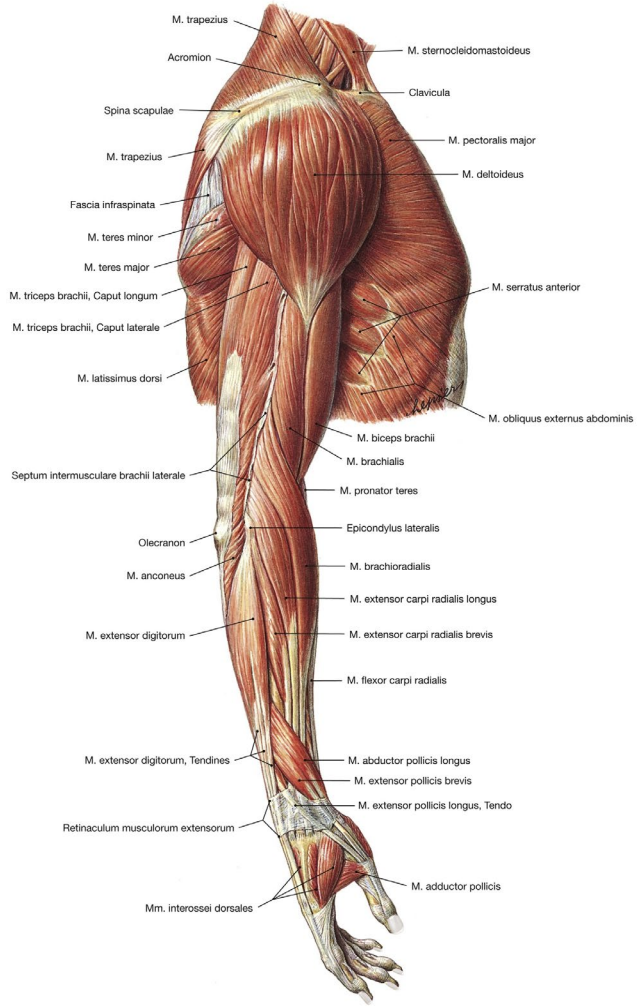
If you know where a muscle lies and which joint it crosses, you can often predict its function.

Muskel → sene → krydser et led → kontraktion → bevægelse

Muscle → tendon → crosses a joint → contraction → movement









Muskelskema

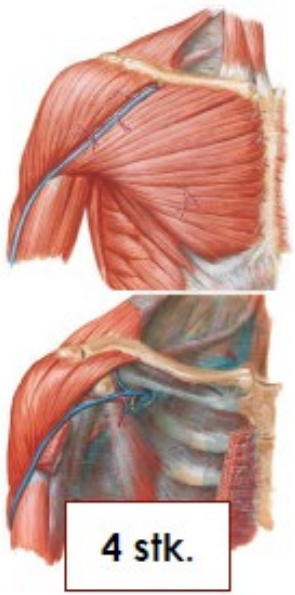
NAVN	UDSPRING	INSERTION	VIRKNING
GRUPPE, *innervationsniveau* OBS: Innervationsniveauerne kan variere 1-2 segmenter.			
1. RYG- OG NAKKEMUSKLER (m.musculi dorsi) – dybe (overfladiske: se 6.1.a)			
1. Musculus erector spinae:			
rr. posteriores n. spinalium C1-S3			
a. M. iliocostalis:	Samlet udspring; fascia thora-columb. os sacrum, crista iliaca, processus transversi L1-5		Extension, lateralflexion og sikring af columnas oprejste stilling
M. iliocostalis lumborum	Crista iliaca	Anguli af 5.-12. ribben	Extension og lateralflexion
M. iliocostalis thoracis	Mediale side af de 6 nederste ribbensanguli	6 øverste ribben(anguli)	Extension og lateralflexion
M. iliocostalis cervicis	Kraniale og midterste ribben	Processus transversi af de midterste halshvirvler	Extension og lateralflexion
b. M. longissimus:	Crista iliaca, os sacrum col. lumb.	Pr. transversi lænde-, bryst- og halshvirvler, ribben medialt for anguli	Extension og lateralflexion
M. longissimus thoracis	Processus transversi af S4-L1, proc. transv. T12-7	Proc. transversi af lænde- og bryst-hvirvlerne, anguli af ribben T12-2	Extension og lateralflexion
M. longissimus cervicis	Proc. transv. T6-1	Proc. transv. C7-2	Extension og lateralflexion
M. longissimus capitis	Proc. transv. af T3-C3	Proc. mastoideus	Extension, lateralflexion og rotation af hovedet
c. M. spinalis:	Proc. spinosi L2 – C6	Proc. spinosi T9-C2	Extension og lateralflexion
M. spinalis thoracis	Proc. spin. af L2-T11	Proc. spin. af T9-2	Extension og lateralflexion
M. spinalis cervicis	Proc. spin. af T2-C6	Proc. spin. af C4-2	Extension
2. Musculi transversospinales:			
rr. posteriores nn. spinalium			
M. semispinalis:		Overspringer 4 el. flere hvirvler	
M. semispinalis thoracis	Proc. transv. af T12-6	Proc. spin. af T6-C6	Extension, rotation til modsat side
M. semispinalis cervicis	Proc. transv. af T6-1	Proc. spin. af C5-2	Extension, rotation til modsat side
M. semispinalis capitis	Proc. transv. af T6-C4	Os occipitale under linea nuch. sup.	Extension, vigtig for hovedets holdning
Mm. multifidi	Proc. transv. L5-1	Proc. spin. – overspringende 2-3 hvirvler.	Lateralflexion af columna og lidt rotation til modsat side
Mm. rotatores	Proc. transv.	Nærmest ovenliggende proc. spin	Rotation til modsat side
3. Musculi interspinales			
4. Musculi intertransversarii			
5. Musculi suboccipitales:			
rr. posteriores C1-C2			
M. rectus capitis posterior major	Proc. spin. C2	Os occipitale under linea nuchalis inf.	Rotation og extension af hovedet
M. rectus capitis posterior minor	Tub. post. C1	Medialt under linea nuchalis inf.	Extension
M. obliquus capitis superior	Proc. transv. C1	Os occipitale lige over insertionen for m. rectus cap. post. major	Extension og lateralflexion
M. obliquus capitis inferior	Proc. spin. C2	Proc.transv. C1	Rotation af atlas og hovedet til samme side

Skulderen

The shoulder

De overfladiske brystmuskler

The superficial chest muscles



- ***M. pectoralis major***
- ***M. pectoralis minor***
- ***M. subclavius***
- ***M. serratus anterior***

De overfladiske rygmuskler

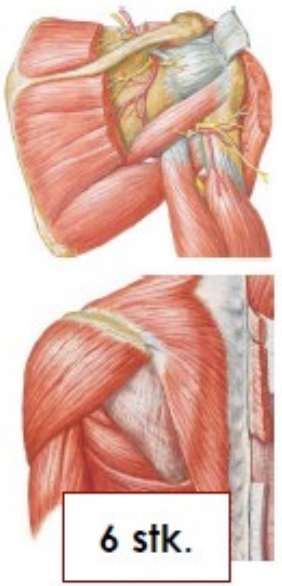
The superficial back muscles



- ***M. trapezius***
- ***M. latissimus dorsi***
- ***M. levator scapulae***
- ***M. rhomboidei***

De egentlige skuldermuskler

The intrinsic shoulder muscles



- ***M. deltoideus***
- ***M. teres major***
- ***M. teres minor***
- ***M. supraspinatus***
- ***M. infraspinatus***
- ***M. subscapularis***

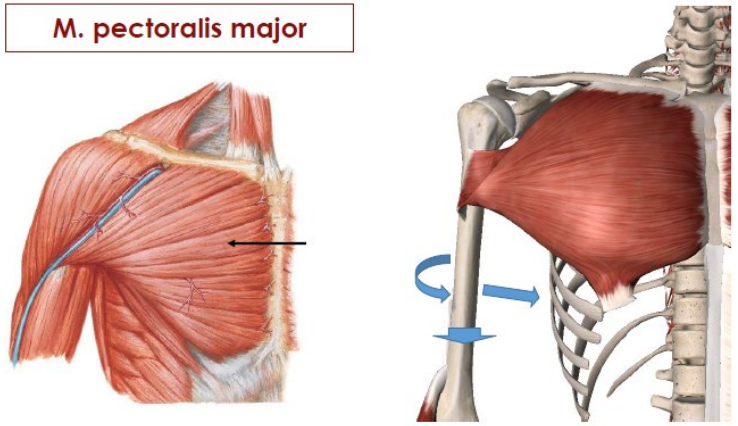
Skulderen

The shoulder

De overfladiske brystmuskler--- 1. lag

The superficial chest muscles--- 1st layer

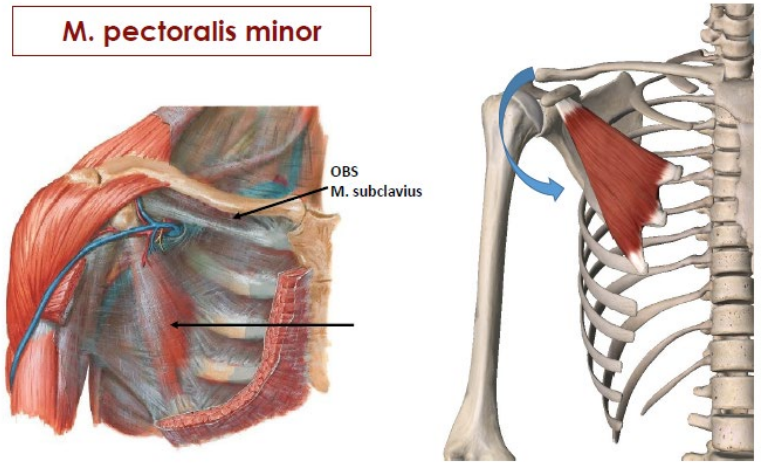
M. pectoralis major



De overfladiske brystmuskler--- 2. lag

The superficial chest muscles--- 2nd layer

M. pectoralis minor

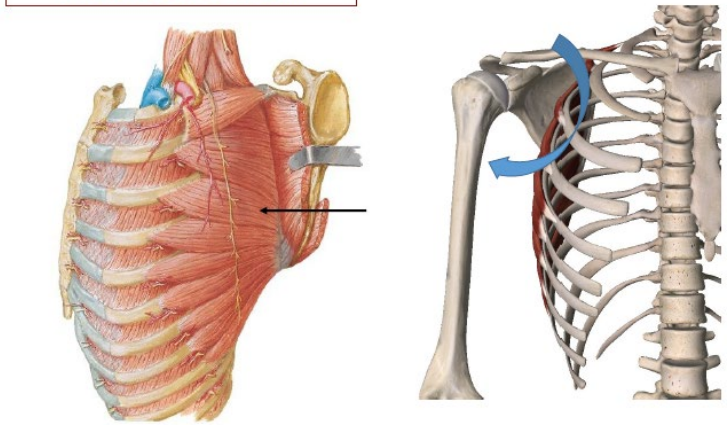


M. subclavius

De overfladiske brystmuskler--- 3. lag

The superficial chest muscles--- 3rd layer

M. serratus anterior



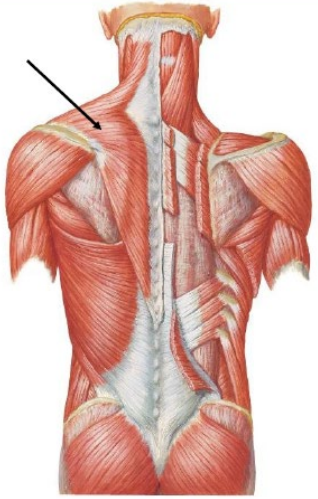


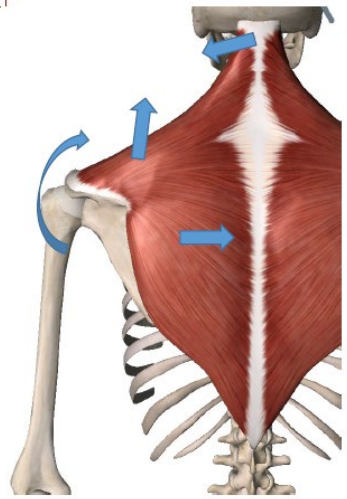
De overfladiske rygmuskler

The superficial back muscles

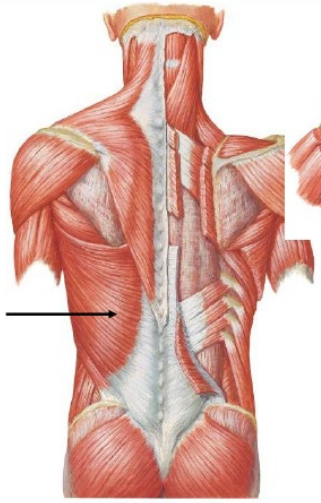


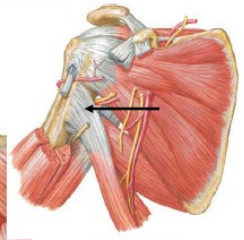
M. trapezius

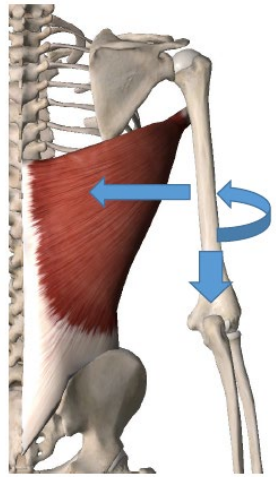




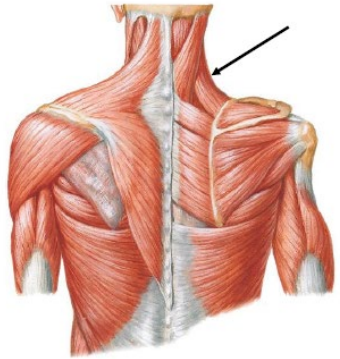
M. latissimus dorsi

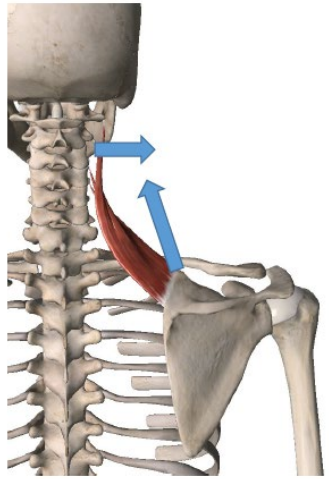




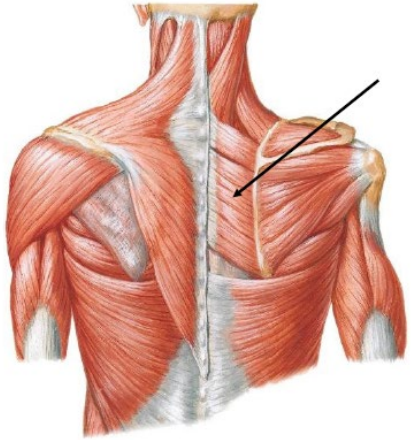


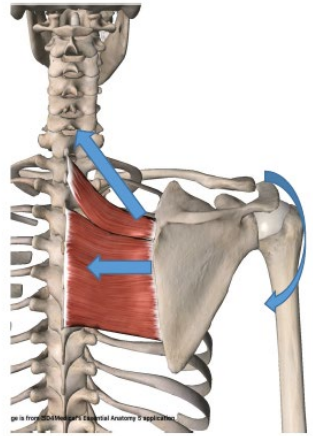
M. levator scapulae





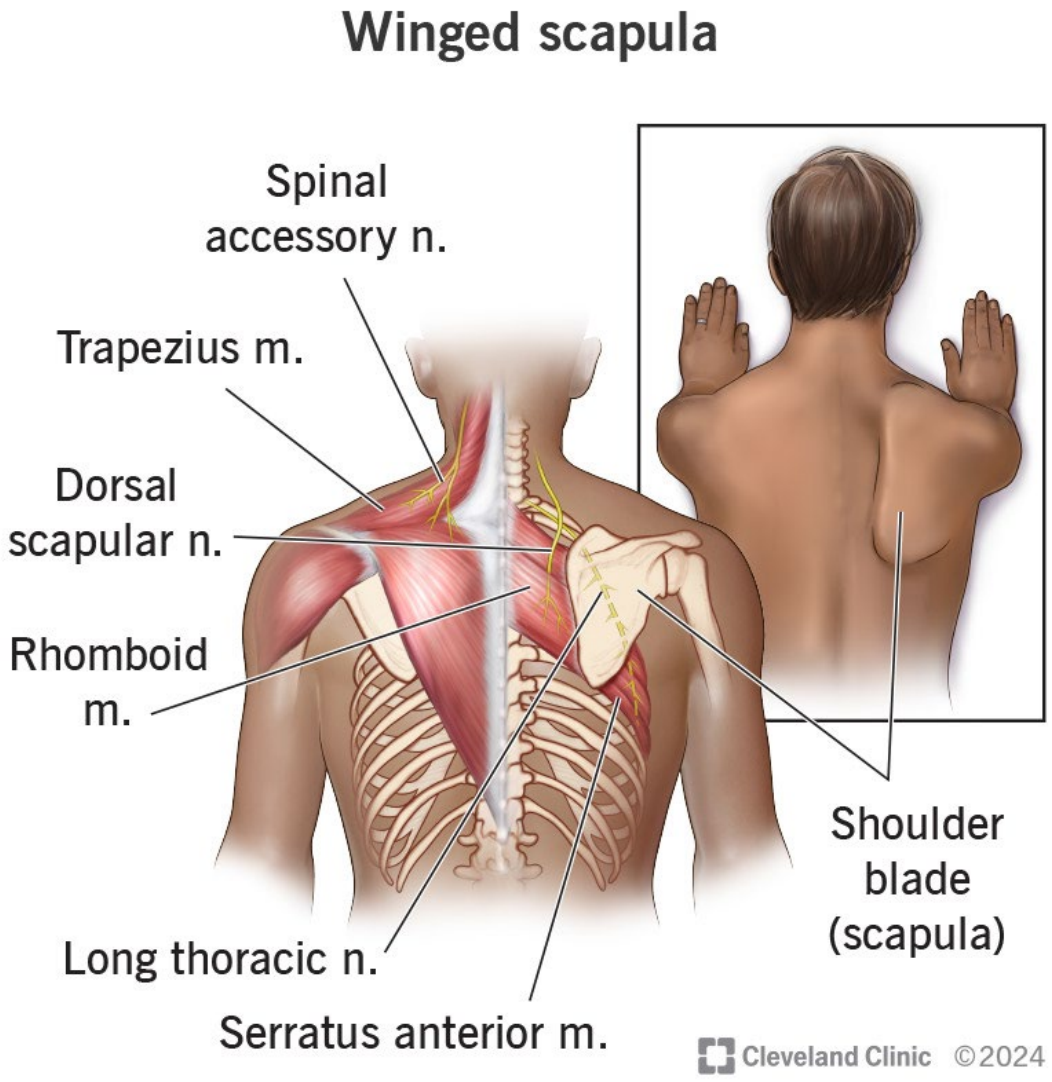
M. rhomboideus





De overfladiske rygmuskler

The superficial back muscles



De egentlige skuldermuskler

The intrinsic shoulder muscles

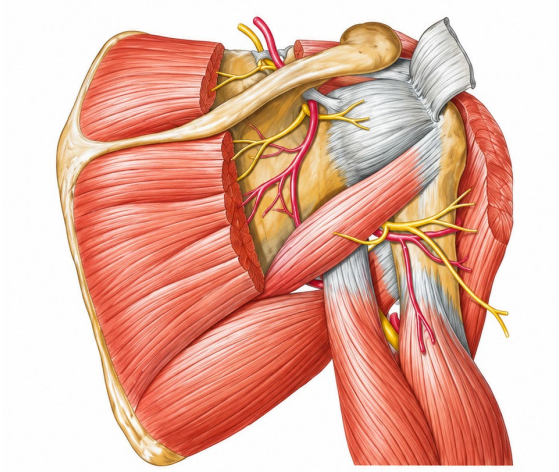
Generelt:

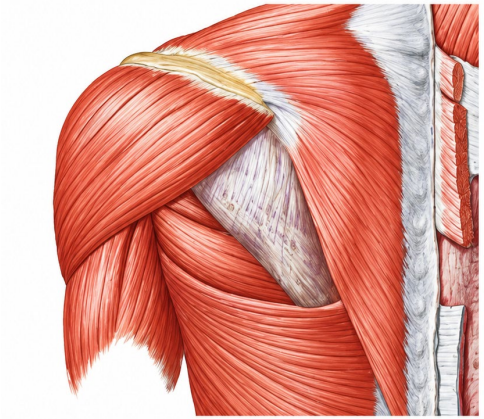
– **To funktioner:**

- Bevæge skulderleddet
  Move the shoulder joint

- Holde ledhovedet stabilt i ledskålen
  Keep the humeral head stable in the glenoid cavity

– Danner **"rotatormanchetten"** omkring skulderleddet
  Form the "rotator cuff" around the shoulder joint
  - Har stor betydning for skulderleddets stabilitet
    Plays an important role in stabilizing the shoulder joint





6 stk.

M. deltoideus

M. teres major

M. teres minor

M. supraspinatus

M. infraspinatus

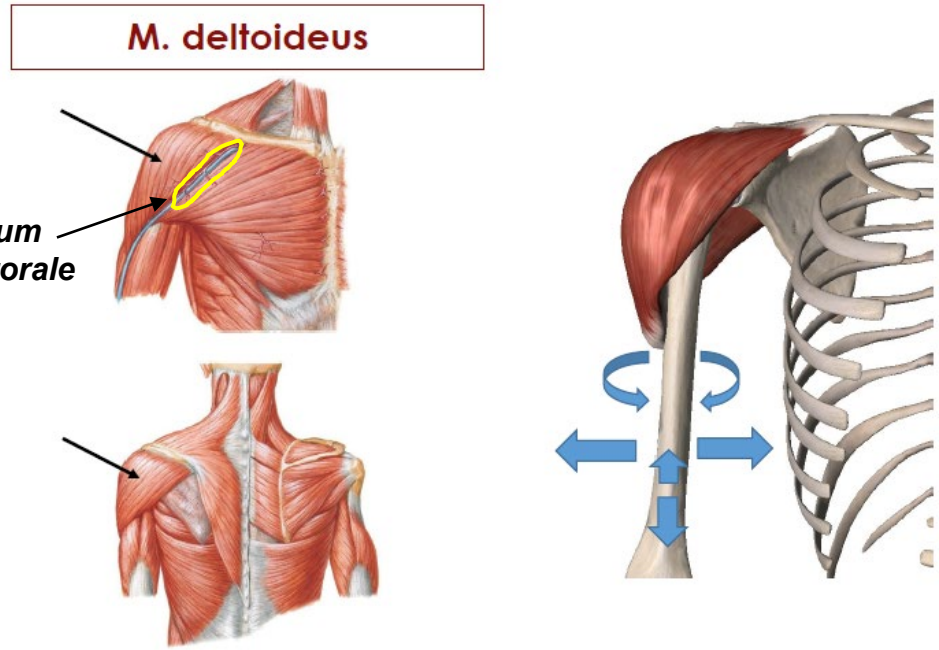
M. subscapularis

De egentlige skuldermuskler
The intrinsic shoulder muscles

M. Deltoideus

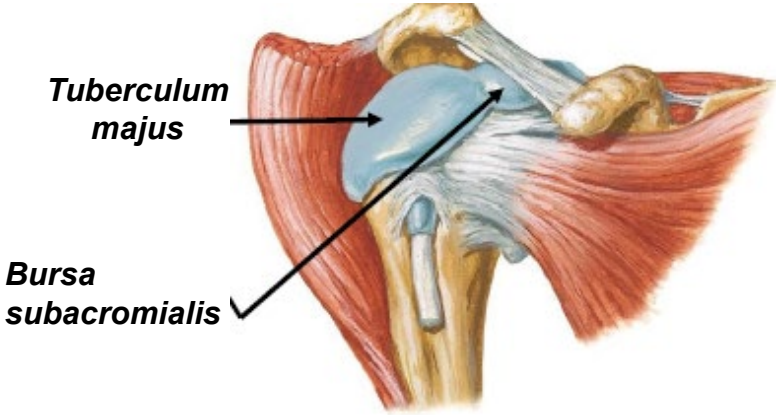
Trigonum clavipectorale
Clavipectoral triangle

- Mellem *m. deltoideus*' forreste fibre og *m. pectoralis major*
  Between the anterior fibers of the deltoid muscle and the pectoralis major muscle
- Indhold:
  - *Processus coracoideus* mærkes i dybden
    The coracoid process can be palpated in the depth
  - *V. cephalica*'s sidste stykke
    Terminal part of the cephalic vein
- Klinisk relevans: Bruges som anatomisk landemærke ved den deltopektorale adgang til skulderleddet.
  Clinical relevance: Used as an anatomical landmark in the deltopectoral approach to the shoulder.



Bursa subdeltoidea
Subdeltoid bursa

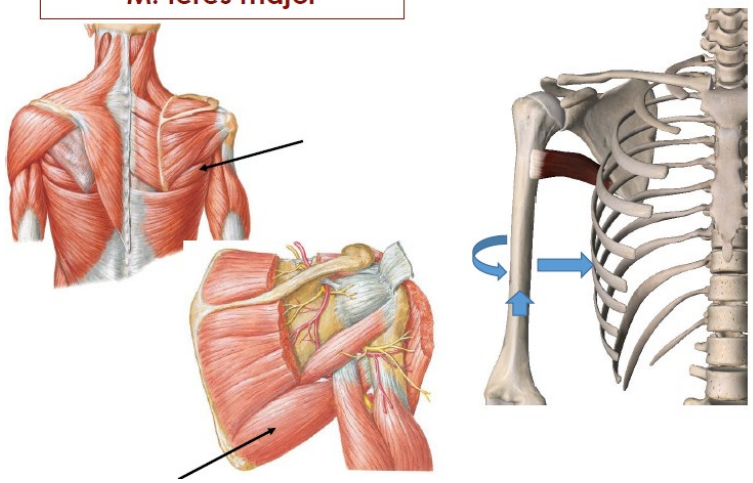
- Under musklen, hvor den glider mod *tuberculum majus*
  Located beneath the muscle where it glides over the greater tubercle
- Kommunikerer med *bursa subacromialis*
  Communicates with the subacromial bursa
- Klinisk relevans: Irritation af den subakromiale/subdeltoide bursa kan give smerter ved abduktion og bevægelser over hovedet.
  Clinical relevance: Irritation of the subacromial/subdeltoid bursa can cause pain during abduction and overhead movement.



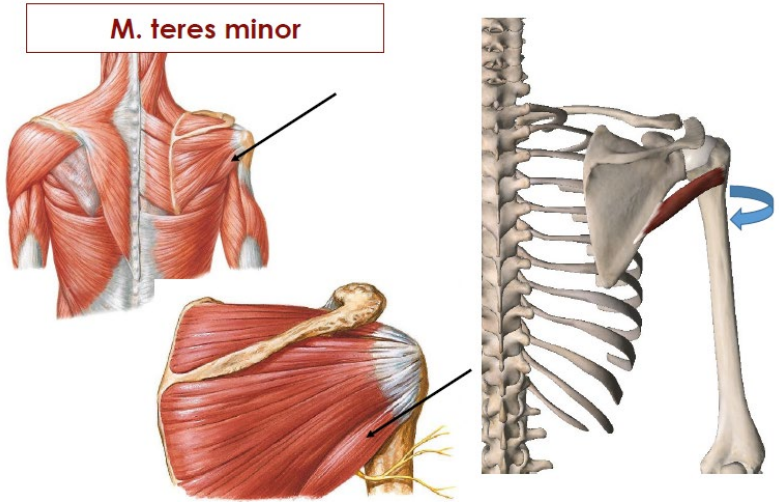
De egentlige skuldermuskler

The intrinsic shoulder muscles

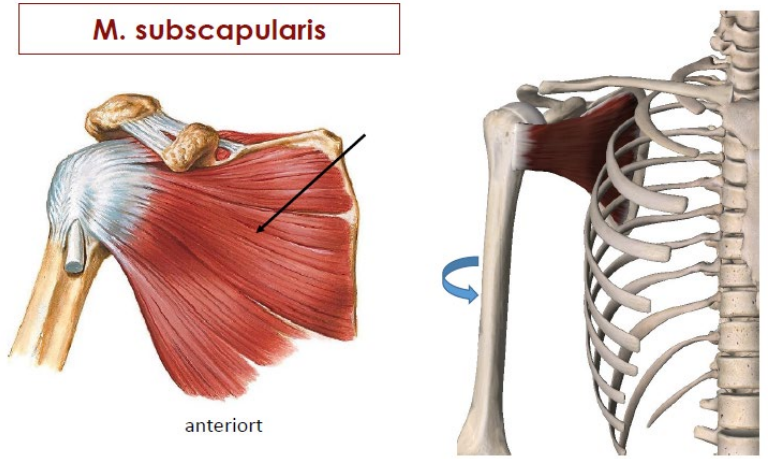
M. teres major



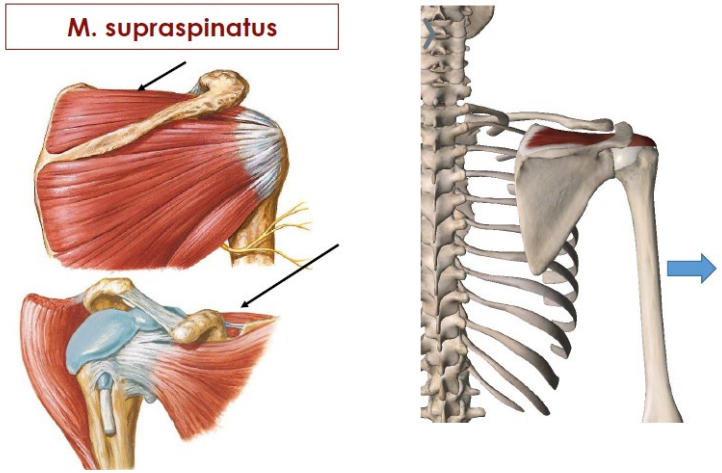
M. teres minor



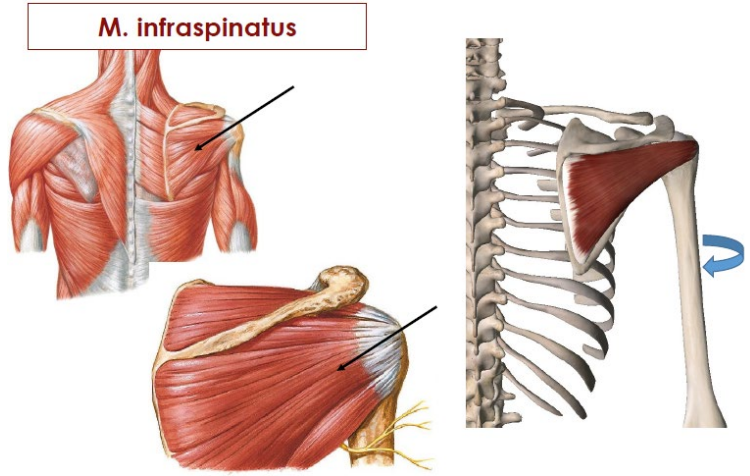
M. subscapularis



M. supraspinatus



M. infraspinatus



De egentlige skuldermuskler

The intrinsic shoulder muscles

Rotatormanchetten
Rotator cuff

Dannet af:

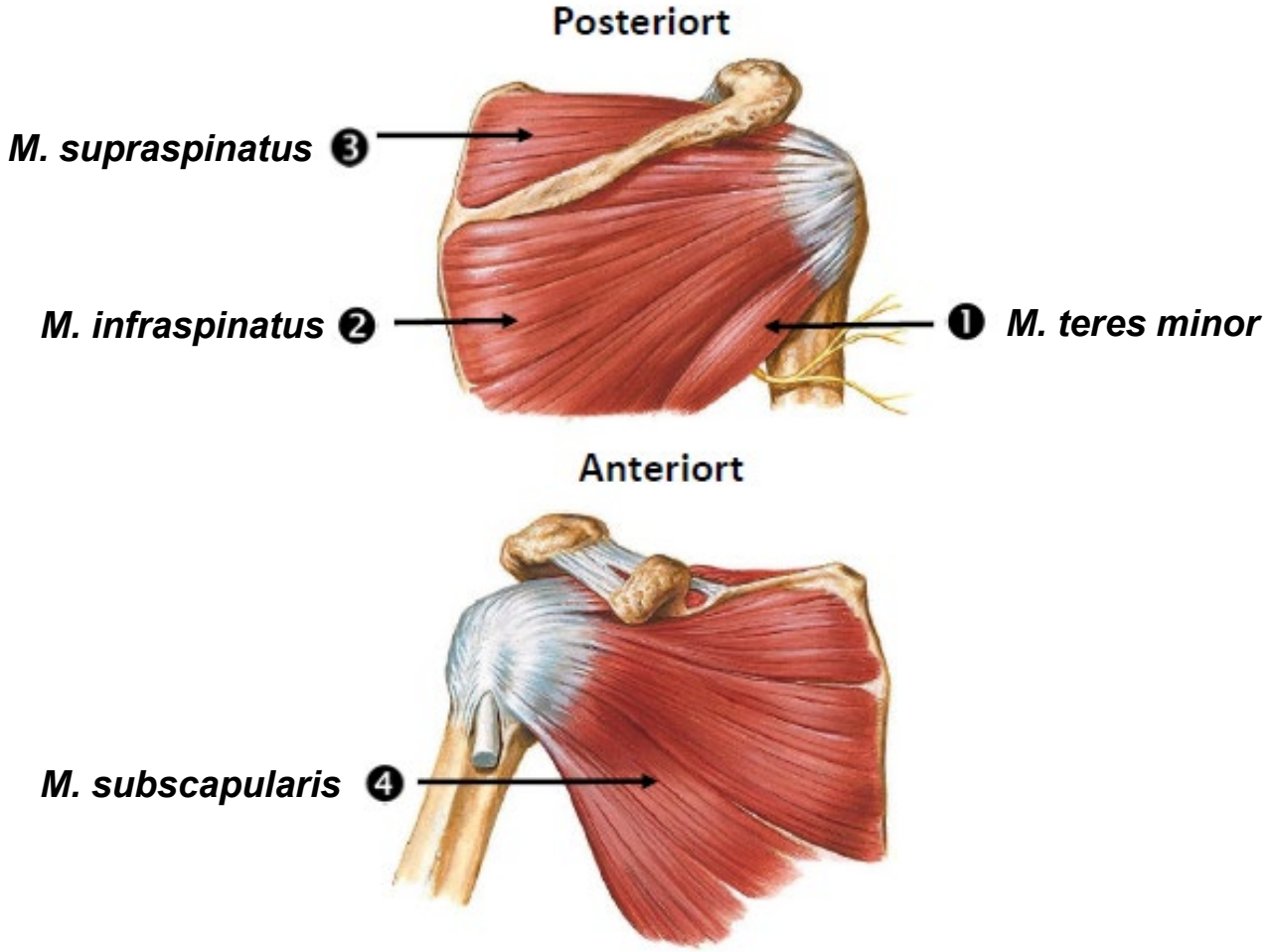
1. *M. teres minor*
2. *M. infraspinatus*
3. *M. supraspinatus*
4. *M. subscapularis*

Hæfter på *tuberculum majus* og *minus*
Attach to the greater and lesser tubercles

Holder *caput humeri* på plads i *cavitas glenoidalis*
Keeps the humeral head in place within the glenoid cavity

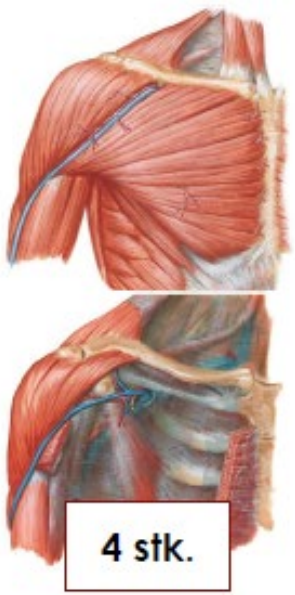
Giver støtte og stabilitet til skulderleddet
Provides support and stability to the shoulder joint

Selvstændig virkning på armen
Each muscle also contributes to specific movements of the arm



Skulderen
The shoulder

De overfladiske brystmuskler
The superficial chest muscles



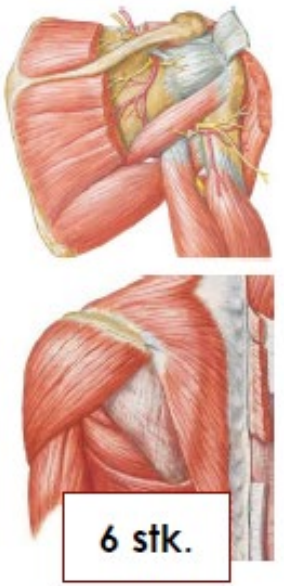
- ***M. pectoralis major***
- ***M. pectoralis minor***
- ***M. subclavius***
- ***M. serratus anterior***

De overfladiske rygmuskler
The superficial back muscles



- ***M. trapezius***
- ***M. latissimus dorsi***
- ***M. levator scapulae***
- ***M. rhomboidei***

De egentlige skuldermuskler
The intrinsic shoulder muscles



- ***M. deltoideus***
- ***M. teres major***
- ***M. teres minor***
- ***M. supraspinatus***
- ***M. infraspinatus***
- ***M. subscapularis***

Rotatormanchetten
Rotator cuff

Læringsmål:

- **Redegøre for de vigtigste knogler og anatomiske landemærker i overekstremiteten**
  Describe the major bones and anatomical landmarks of the upper extremity

- **Redegøre for de vigtigste led og ligamenter samt forklare deres betydning for bevægelse og stabilitet**
  Describe the major joints and ligaments and explain their role in movement and stability

- **Redegøre for de vigtigste muskelgrupper omkring skulderbæltet og skulderleddet samt forklare deres rolle i bevægelse og stabilisering**
  Describe the major muscle groups of the shoulder girdle and shoulder joint and explain their role in movement and stabilization

- **Redegøre for de vigtigste muskelgrupper i overarm og underarm og relatere deres anatomiske placering til deres funktion**
  Describe the major muscle groups of the arm and forearm and relate their anatomical position to their function

- **Redegøre for de vigtigste strukturer i håndled og hånd, herunder fleksorsener, canalis carpi og funktionelt vigtige håndmuskler**
  Describe the major structures of the wrist and hand, including the flexor tendons and functionally important hand muscles

Brachium (overarmen)
Upper arm

Brachiums fleksorer
Flexors of the arm

- ***M. biceps brachii***
  Udspring:
  1. *Caput longum*
     Long head
  2. *Caput breve*
     Short head

  Hovedfunktion: fleksion i albueleddet og supination af underarmen
  Main function: flexion of the elbow and supination of the forearm

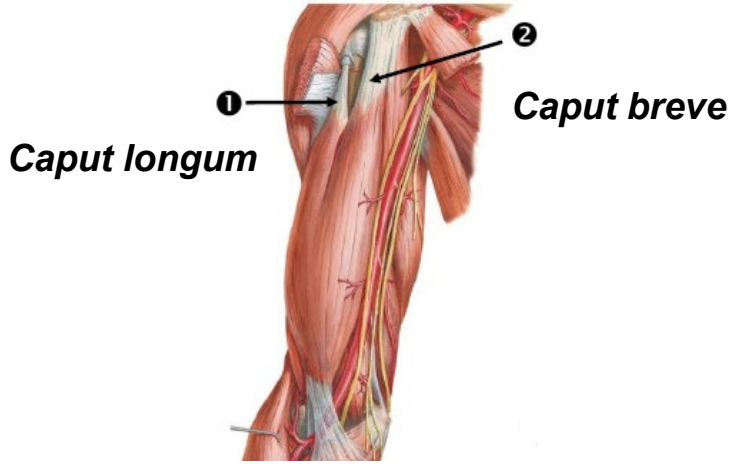
- ***M. coracobrachialis***
  Hovedfunktion: fleksion og adduktion af armen i skulderleddet
  Main function: flexion and adduction of the arm at the shoulder joint

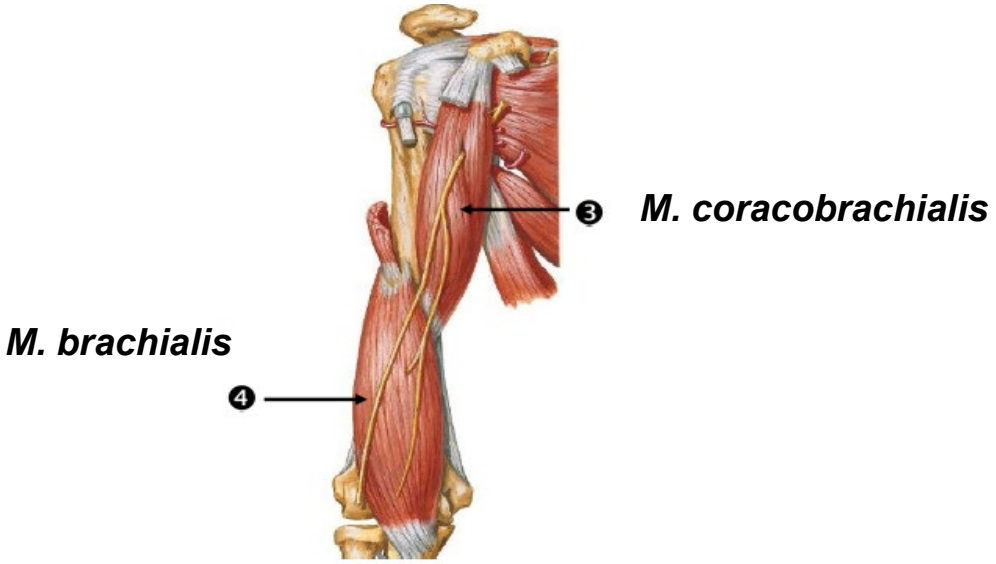
- ***M. brachialis***
  Hovedfunktion: kraftig fleksion i albueleddet
  Main function: powerful flexion of the elbow

Forreste muskelloge i overarmen → primært fleksion
Anterior compartment of the arm → primarily flexion



M. biceps brachii

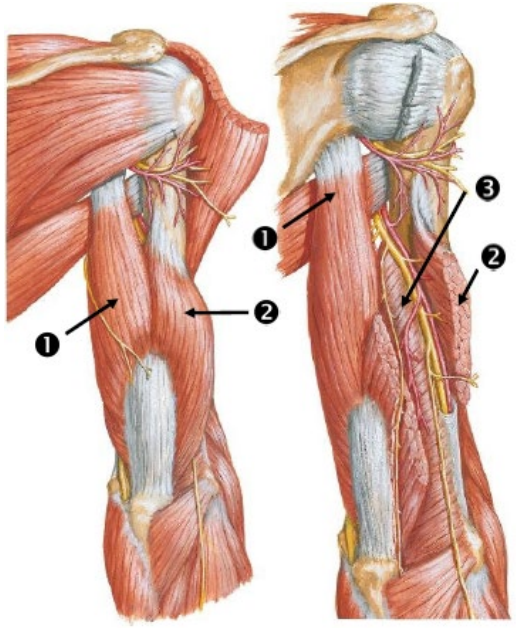




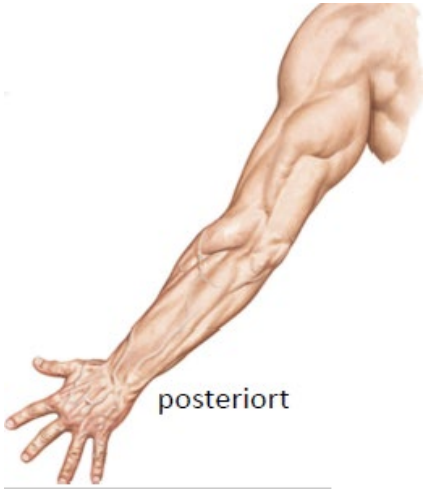
Brachium (overarmen)
Upper arm

Brachiums ekstensorer
Extensors of the arm

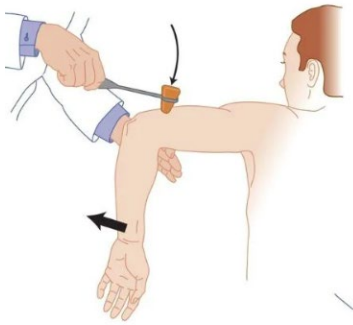
- ***M. triceps brachii***
  Udspring:
  1. *Caput longum* (superficiel)
     Long head
  2. *Caput laterale (superficiel)*
     Short head
  3. *Caput mediale* (profund)
     Medial head



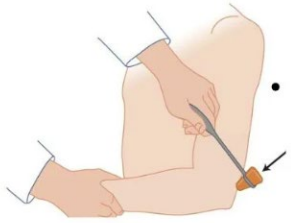
posteriort



The triceps jerk is a deep tendon reflex used in neurological exams **to assess the function of the C7 and C8 spinal nerve roots**, primarily through the radial nerve.



Patient Position:
- Arm should be partially flexed at the elbow and completely relaxed.
- The forearm can be supported by the examiner or rest on the patient's lap if seated.

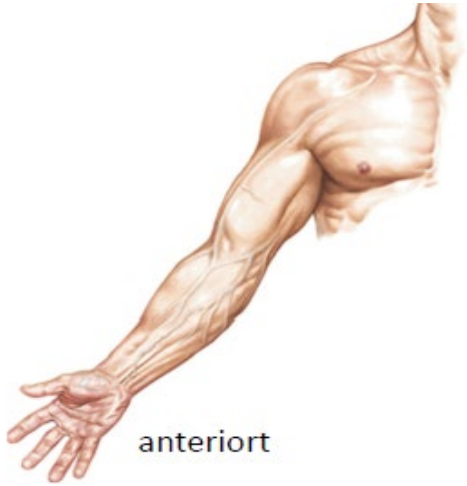


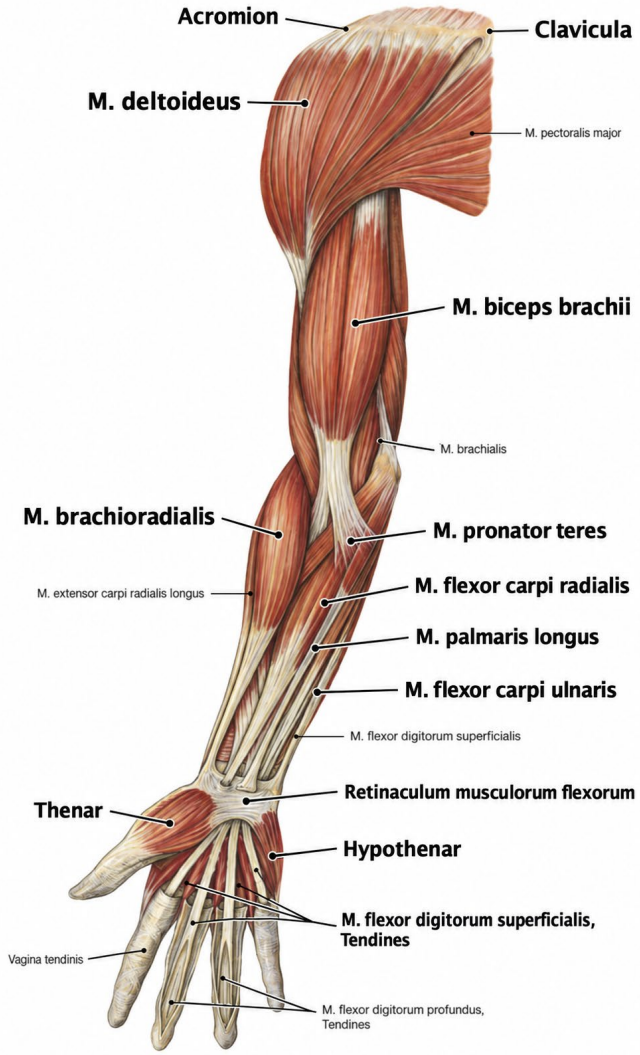
Examiner Action:
- Locate the triceps tendon just above **the olecranon process (elbow)**.
- Tap the tendon directly with a reflex hammer.

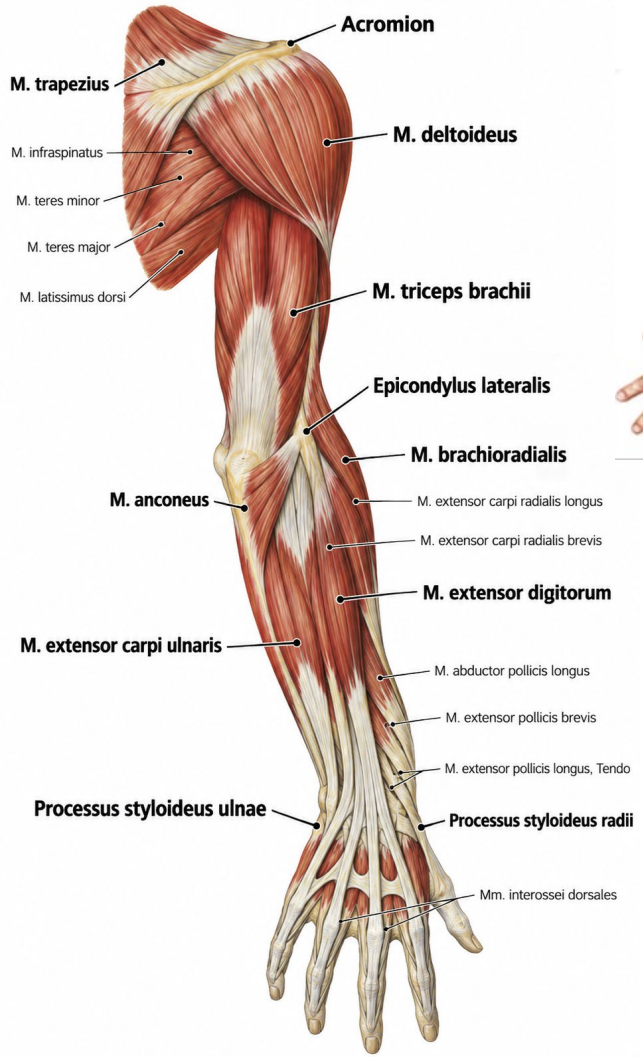
Triceps Reflex

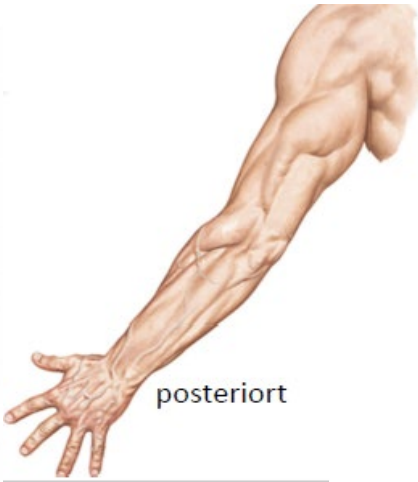
Find de overfladiske anatomiske landemærker

Find the surface anatomical landmarks







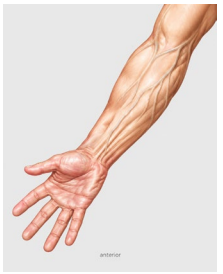


Antebrachium (Underarmen)
Forearm

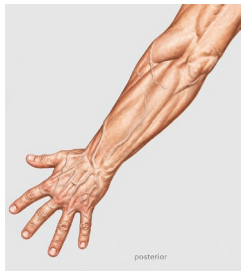
Muskler proksimalt og sener distalt
Muscles proximally and tendons distally

Sikrer, at hånd og fingre er lette og bevægelige.
Helps keep the hand and fingers light and mobile.

- Ventralt: fleksorloge
  Anteriorly: flexor compartment
  - Superficielle muskler → epicondylus medialis humeri
    Superficial muscles → medial epicondyle of the humerus
  - Profunde muskler → forfladen af knoglerne og membrana interossea
    Deep muscles → anterior surfaces of the forearm bones and the interosseous membrane
  - 4 lag muskler: 4, 1, 2 og 1
    Four muscular layers: 4, 1, 2 and 1 muscles



- Dorsalt: ekstensorloge
  Posteriorly: extensor compartment
  - Superficielle muskler → epicondylus lateralis humeri
    Superficial muscles → lateral epicondyle of the humerus
  - Profunde muskler → bagfladen af knoglerne og membrana interossea
    Deep muscles → posterior surfaces of the forearm bones and the interosseous membrane
  - 3 lag muskler: 3, 4 og 5
    Three muscular layers: 3, 4 and 5 muscles



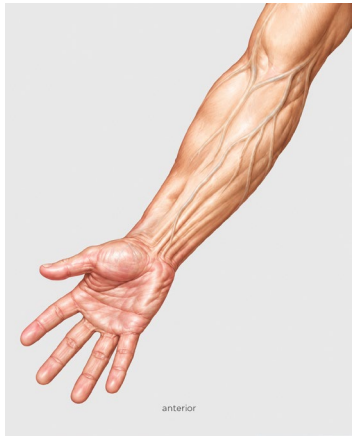


Antebrachium (Underarmen)

Forearm

Ventralt: fleksorloge

Anteriorly: flexor compartment



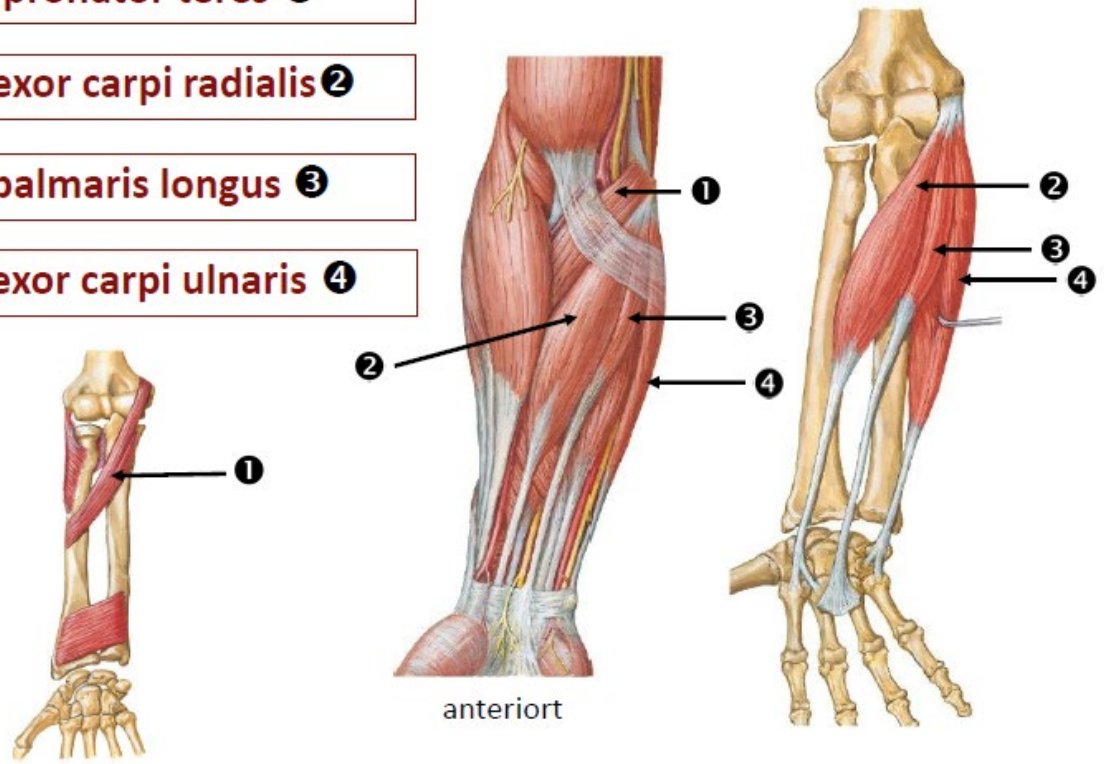
Underarmens fleksorer 1. lag

M. pronator teres ❶

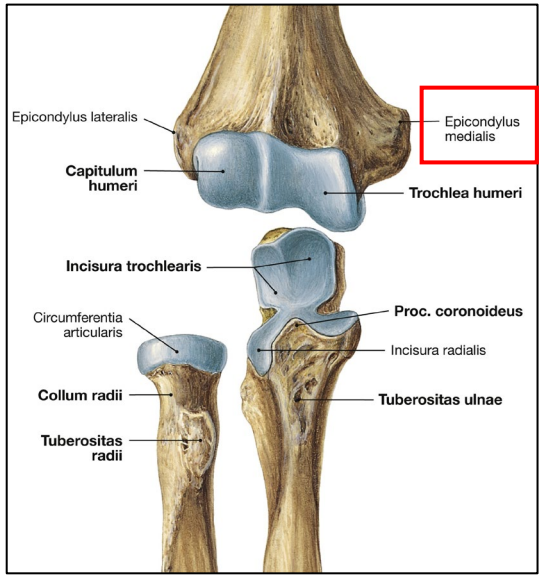
M. flexor carpi radialis ❷

M. palmaris longus ❸

M. flexor carpi ulnaris ❹



Epicondylus medialis humeri

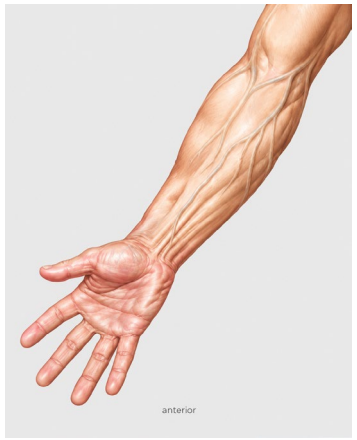


Antebrachium (Underarmen)

Forearm

Ventralt: fleksorloge

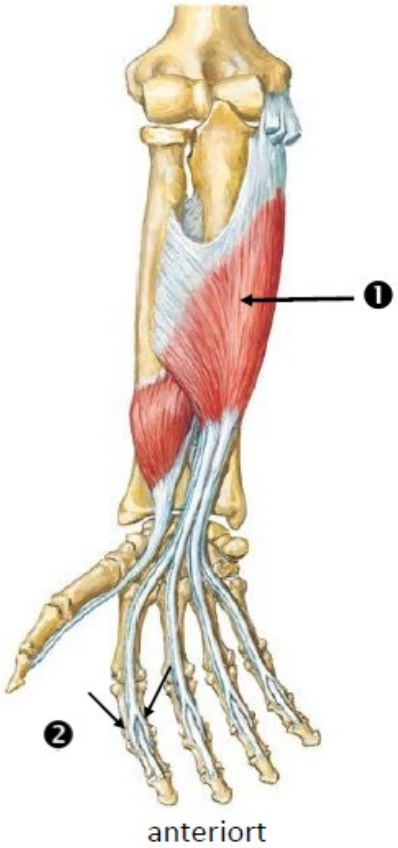
Anteriorly: flexor compartment



Underarmens fleksorer 2. lag

M. flexor digitorum superficialis ❶

- Igennem canalis carpi
- Spalter sig i 2 snipper, der insererer på ❷ hver side af phalanx media II-V



Fleksion af især PIP-leddene i finger II–V

Primarily flexes the PIP joints of digits II–V

PIP-leddene: proksimale interfalangealled

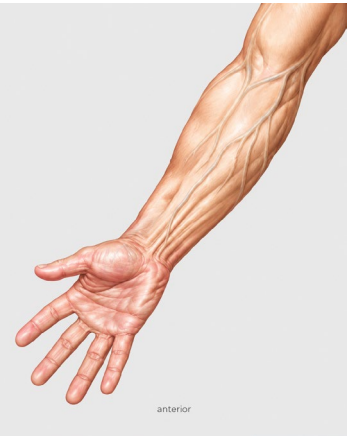
proximal interphalangeal joints

Antebrachium (Underarmen)

Forearm

Ventralt: fleksorloge

Anteriorly: flexor compartment



Underarmens fleksorer 3. + 4. lag

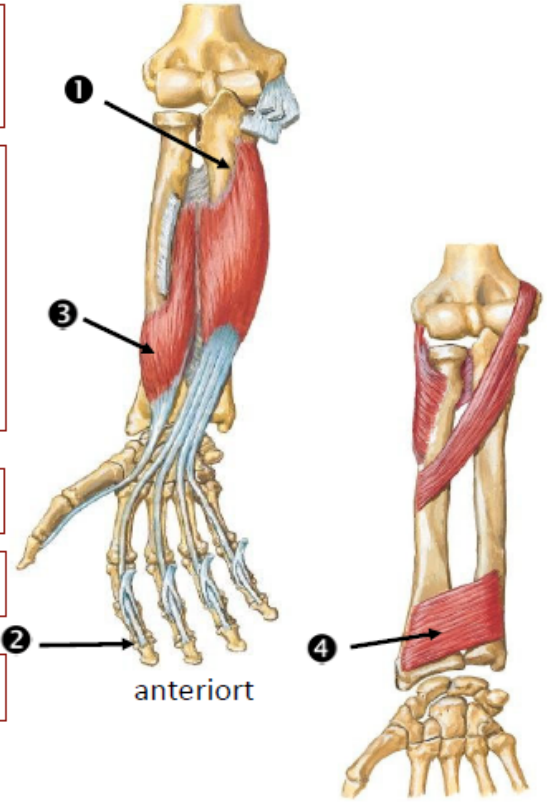
❶ M. flexor digitorum profundus

- Igennem canalis carpi
- Løber profund for superficialissenerne igennem slidsen ud for phalanx media og inserer på ❷ basis af phalanx distalis II-V

❸ M. flexor pollicis longus

- Igennem canalis carpi

❹ M. pronator quadratus



Antebrachium (Underarmen)

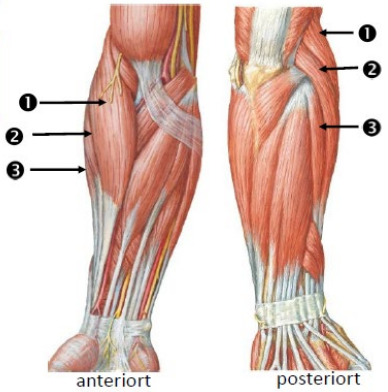
Forearm

Underarmens ekstensorer 1. lag

❶ M. brachioradialis

❷ M. extensor carpi radialis longus

❸ M. extensor carpi radialis brevis





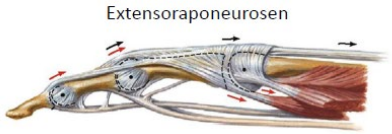
Underarmens ekstensorer 2. lag

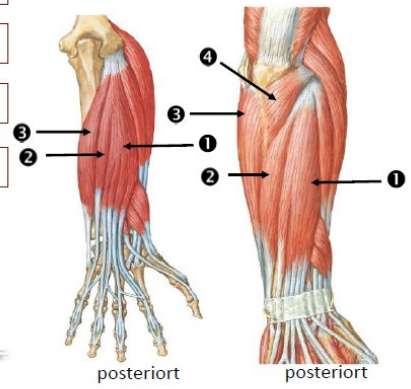
❶ M. extensor digitorum

❷ M. extensor digiti minimi

❸ M. extensor carpi ulnaris

❹ M. anconeus





Underarmens ekstensorer 3. lag

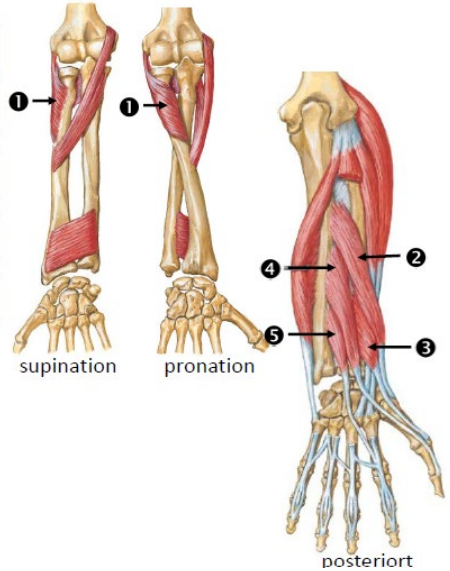
❶ M. supinator

❷ M. abductor pollicis longus

❸ M. extensor pollicis brevis

❹ M. extensor pollicis longus

❺ M. extensor indicis



Læringsmål:

- **Redegøre for de vigtigste knogler og anatomiske landemærker i overekstremiteten**
  Describe the major bones and anatomical landmarks of the upper extremity

- **Redegøre for de vigtigste led og ligamenter samt forklare deres betydning for bevægelse og stabilitet**
  Describe the major joints and ligaments and explain their role in movement and stability

- **Redegøre for de vigtigste muskelgrupper omkring skulderbæltet og skulderleddet samt forklare deres rolle i bevægelse og stabilisering**
  Describe the major muscle groups of the shoulder girdle and shoulder joint and explain their role in movement and stabilization

- **Redegøre for de vigtigste muskelgrupper i overarm og underarm og relatere deres anatomiske placering til deres funktion**
  Describe the major muscle groups of the arm and forearm and relate their anatomical position to their function

- **Redegøre for de vigtigste strukturer i håndled og hånd, herunder fleksorsener, canalis carpi og funktionelt vigtige håndmuskler**
  Describe the major structures of the wrist and hand, including the flexor tendons and functionally important hand muscles

Håndled
wrist

Fascia antebrachii – forstærkning
Reinforcement of the antebrachial fascia



Retinaculum extensorum
Extensor retinaculum

1. kulisse:
First extensor compartment
- ***M. abductor pollicis longus***
  Abductor pollicis longus muscle
- ***M. extensor pollicis brevis***
  Extensor pollicis brevis muscle

2. kulisse:
Second extensor compartment
- ***M. extensor carpi radialis longus***
  Extensor carpi radialis longus muscle
- ***M. extensor carpi radialis brevis***
  Extensor carpi radialis brevis muscle

3. kulisse
Third extensor compartment
- ***M. extensor pollicis longus***
  Extensor pollicis longus muscle

4. kulisse
Fourth extensor compartment
- ***M. extensor digitorum***
  Extensor digitorum muscle
- ***M. extensor indicis***
  Extensor indicis muscle

5. kulisse
Fifth extensor compartment
- ***M. extensor digiti minimi***
  Extensor digiti minimi muscle

6. kulisse
Sixth extensor compartment
- ***M. extensor carpi ulnaris***
  Extensor carpi ulnaris muscle

Håndled
wrist

Tabatièren
Anatomical snuffbox

Afgrænses af
Boundaries

- **Senerne fra 1. kulisse:**
  Tendons of the first extensor compartment
  - *M. abductor pollicis longus*
    Abductor pollicis longus muscle
  - *M. extensor pollicis brevis*
    Extensor pollicis brevis muscle
- **Senen fra 3. kulisse:**
  Tendon of the third extensor compartment
  - *M. extensor pollicis longus*
    Extensor pollicis longus muscle

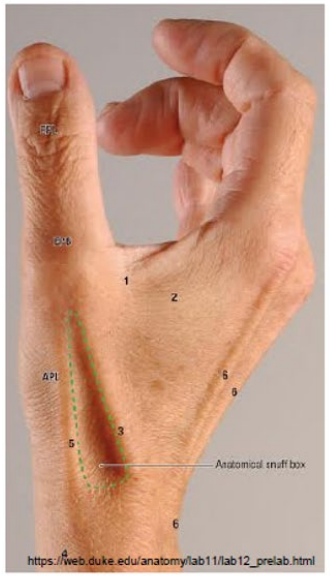
Indhold i tabatièren
Contents of the anatomical snuffbox

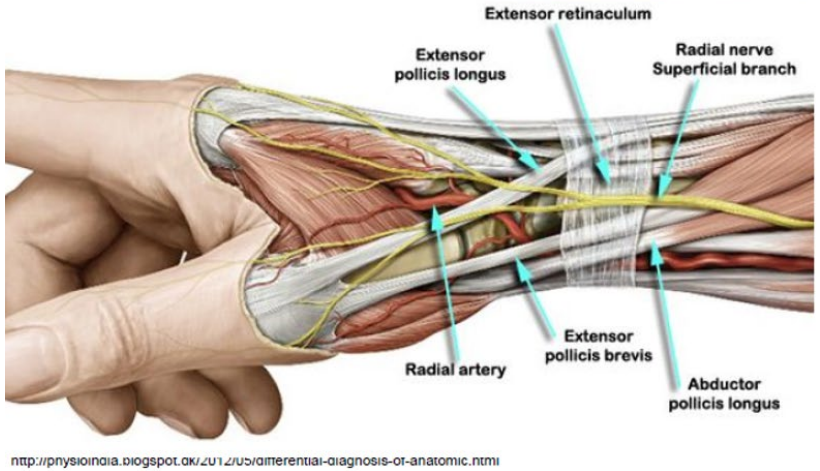
- *A. radialis*
  Radial artery
- *Os scaphoideum*
  Scaphoid bone
- *Processus styloideus radii*
  Radial styloid process

Klinik:
Fraktur af *os scaphoideum*
Scaphoid fracture

- **Vigtigt ikke at overse pga. blodforsyning og håndens funktion**
  Important not to miss because of the scaphoid's blood supply and its importance for hand function



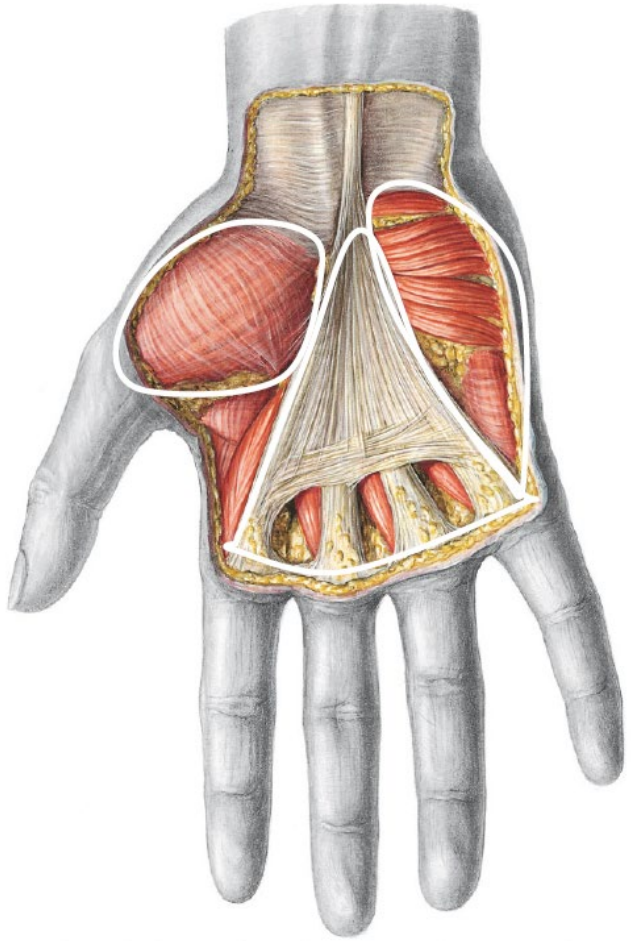




Hånd
Hand

Områder
Regions

- Thenar: tommelfingerbalden
  Thenar eminence
- Hypothenar: lillefingerbalden
  Hypothenar eminence
- Håndens centrale loge: området mellem thenar og hypothenar
  Central compartment of the hand: the region between the thenar and hypothenar compartments
- *Palma manus*: håndflade
  Palm of the hand
- *Dorsum manus*: håndryg
  Dorsum of the hand
- *Canalis carpi*: indgangen til den centrale loge
  Carpal tunnel: the entrance to the central compartment of the hand



Hånd
Hand

Håndens muskler
Muscles of the hand

Thenars muskler

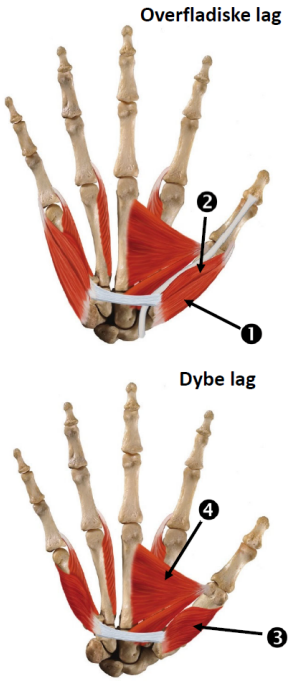
❶ M. abductor pollicis brevis

❷ M. flexor pollicis brevis

❸ M. opponens pollicis

❹ M. adductor pollicis

4 muskler – thumb



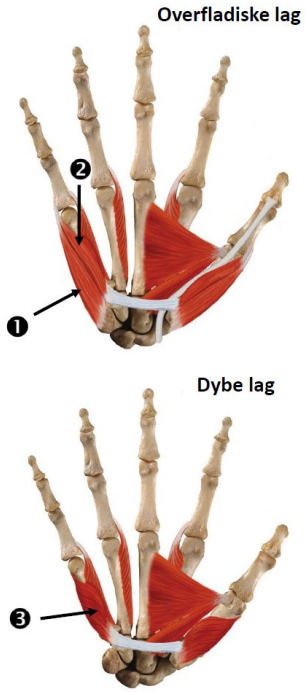
Hypothenars muskler

❶ M. abductor digiti minimi

❷ M. flexor digiti minimi

❸ M. opponens digiti minimi

3 muskler – 5. finger



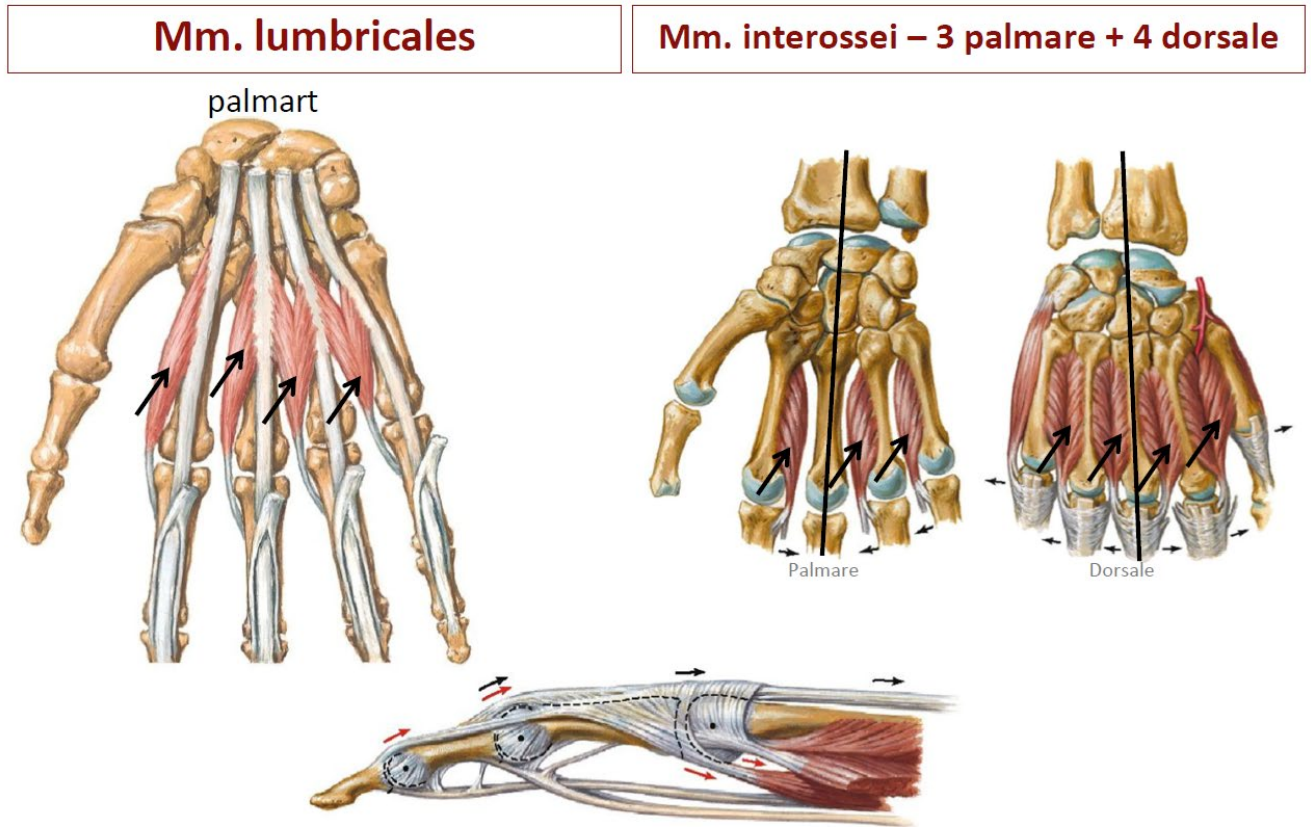
Hånd
Hand

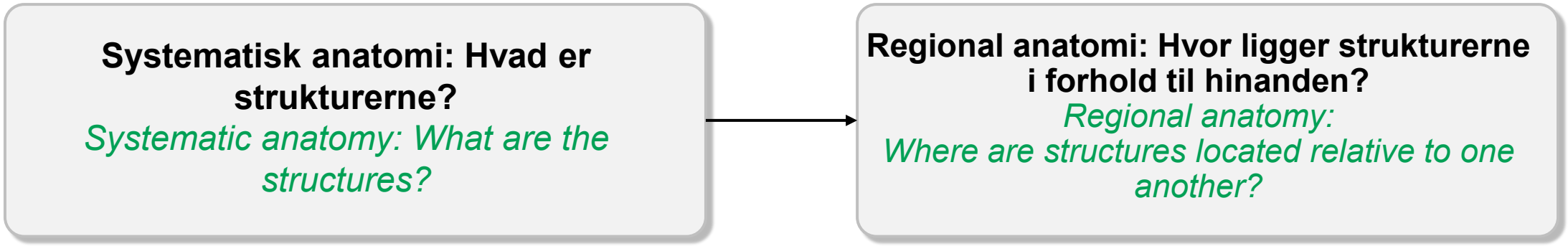
Håndens muskler
Muscles of the hand

Den centrale muskelgruppe
Central muscle group

- ***Mm. lumbricales***
  Lumbrical muscles
  - **4 muskler**

- ***Mm. interossei***
  Interosseous muscles
  - **7 muskler**




Systematisk anatomi: Hvad er strukturerne?
Systematic anatomy: What are the structures?
Regional anatomi: Hvor ligger strukturerne i forhold til hinanden?
Regional anatomy:
Where are structures located relative to one another?